官方示范： 可以在这里查询图列和代码 ：http://matplotlib.sourceforge.net/gallery.html#

Python图表绘制：matplotlib绘图库入门

matplotlib 是 python 最著名的绘图库，它提供了一整套和 matlab 相似的命令 API，十分适合交互式地行制图。而且也可以方便地将它作为绘图控件，嵌入 GUI 应用程序中。

它的文档相当完备，并且 Gallery 页面中有上百幅缩略图，打开之后都有源程序。因此如果你需要绘制某种类型的图，只需要在这个页面中浏览/复制/粘贴一下，基本上都能搞定。

在 Linux 下比较著名的数据图工具还有 gnuplot，这个是免费的，Python 有一个包可以调用 gnuplot，但是语法比较不习惯，而且画图质量不高。

而 Matplotlib则比较强：Matlab的语法、python语言、latex的画图质量（还可以使用内嵌的latex引擎绘制的数学公式）。

Matplotlib.pyplot 快速绘图

快速绘图和面向对象方式绘图

matplotlib 实际上是一套面向对象的绘图库，它所绘制的图表中的每个绘图元素，例如线条 Line2D、文字 Text、刻度等在内存中都有一个对象与之对应。

为了方便快速绘图 matplotlib 通过 pyplot 模块提供了一套和 MATLAB 类似的绘图 API，将众多绘图对象所构成的复杂结构隐藏在这套 API 内部。我们只需要调用 pyplot 模块所提供的函数就可以实现快速绘图以及设置图表的各种细节。pyplot 模块虽然用法简单，但不适合在较大的应用程序中使用。

为了将面向对象的绘图库包装成只使用函数的调用接口，pyplot 模块的内部保存了当前图表以及当前子图等信息。当前的图表和子图可以使用 plt.gcf()和 plt.gca()获得，分别表示"Get Current Figure"和"Get Current Axes"。在 pyplot 模块中，许多函数都是对当前的 Figure 或 Axes 对象进行处理，比如说：

> plt.plot()实际上会通过plt.gca()获得当前的Axes对象ax，然后再调用ax.plot()
>
> 方法实现真正的绘图。

可以在 Ipython 中输入类似"plt.plot??"的命令查看 pyplot 模块的函数是如何对各种绘图对象进行包装的。

配置属性

matplotlib 所绘制的图表的每个组成部分都和一个对象对应，我们可以通过调用这些对象的属性设置方法 set_*()或者 pyplot 模块的属性设置函数 setp()设置它们的属性值。

因为 matplotlib 实际上是一套面向对象的绘图库，因此也可以直接获取对象的属性

配置文件

绘制一幅图需要对许多对象的属性进行配置，例如颜色、字体、线型等等。我们在绘图时，并没有逐一对这些属性进行配置，许多都直接采用了 matplotlib 的缺省配置。

matplotlib 将这些缺省配置保存在一个名为“matplotlibrc”的配置文件中，通过修改配置文件，我们可以修改图表的缺省样式。配置文件的读入可以使用 rc_params()，它返回一个配置字典；在 matplotlib 模块载入时会调用 rc_params()，并把得到的配置字典保存到 rcParams 变量中；matplotlib 将使用 rcParams 字典中的配置进行绘图；用户可以直接修改此字典中的配置，所做的改变会反映到此后创建的绘图元素。

绘制多子图（快速绘图）

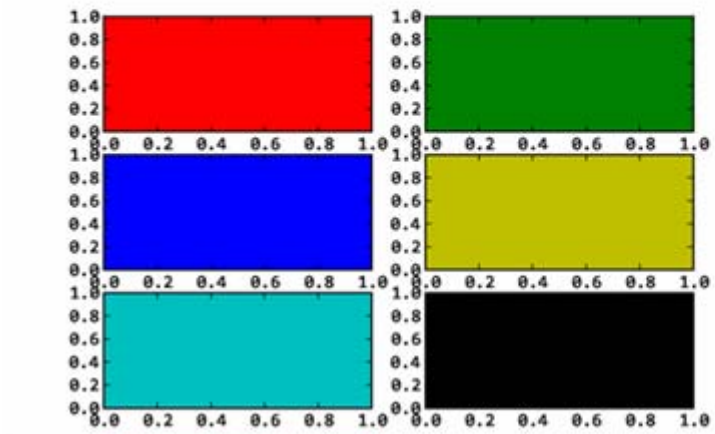
Matplotlib 里的常用类的包含关系为 `Figure -> Axes -> (Line2D, Text, etc.)`一个 Figure 对象可以包含多个子图(Axes)，在 matplotlib 中用 Axes 对象表示一个绘图区域，可以理解为子图。

可以使用 subplot()快速绘制包含多个子图的图表，它的调用形式如下：

```
subplot(numRows, numCols, plotNum)
```

subplot 将整个绘图区域等分为 numRows 行* numCols 列个子区域，然后按照从左到右，从上到下的顺序对每个子区域进行编号，左上的子区域的编号为 1。如果 numRows，numCols 和 plotNum 这三个数都小于 10 的话，可以把它们缩写为一个整数，例如 subplot(323)和 subplot(3,2,3)是相同的。subplot 在 plotNum 指定的区域中创建一个轴对象。如果新创建的轴和之前创建的轴重叠的话，之前的轴将被删除。

```
for idx, color in enumerate("rgbyck"):
    plt.subplot(321+idx, axisbg=color)
plt.show()
```



subplot()返回它所创建的 Axes 对象，我们可以将它用变量保存起来，然后用 sca()交替让它们成为当前 Axes 对象，并调用 plot()在其中绘图。

绘制多图表（快速绘图）

如果需要同时绘制多幅图表，可以给 figure()传递一个整数参数指定 Figure 对象的序号，如果序号所指定的 Figure 对象已经存在，将不创建新的对象，而只是让它成为当前的 Figure 对象。

```
import numpy as np
import matplotlib.pyplot as plt

plt.figure(1) # 创建图表 1

plt.figure(2) # 创建图表 2

ax1 = plt.subplot(211) # 在图表 2 中创建子图 1

ax2 = plt.subplot(212) # 在图表 2 中创建子图 2

x = np.linspace(0, 3, 100)
for i in xrange(5):

    plt.figure(1)  #❶ # 选择图表 1

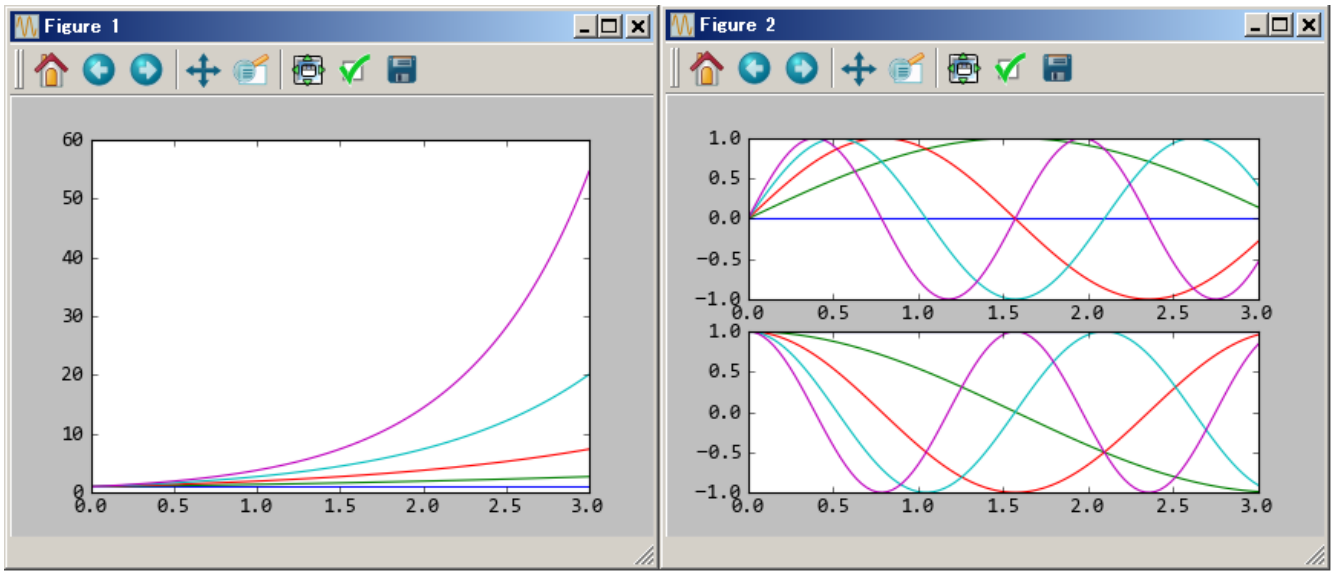
    plt.plot(x, np.exp(i*x/3))

    plt.sca(ax1)   #❷ # 选择图表 2 的子图 1

    plt.plot(x, np.sin(i*x))

    plt.sca(ax2)  # 选择图表 2 的子图 2

    plt.plot(x, np.cos(i*x))
plt.show()
```



在图表中显示中文

matplotlib 的缺省配置文件中所使用的字体无法正确显示中文。为了让图表能正确显示中文，可以有几种解决方案。

1. 在程序中直接指定字体。
2. 在程序开头修改配置字典 rcParams。
3. 修改配置文件。

面向对象画图

matplotlib API 包含有三层，Artist 层处理所有的高层结构，例如处理图表、文字和曲线等的绘制和布局。通常我们只和 Artist 打交道，而不需要关心底层的绘制细节。

直接使用 Artists 创建图表的标准流程如下：

- 创建 Figure 对象
- 用 Figure 对象创建一个或者多个 Axes 或者 Subplot 对象
- 调用 Axies 等对象的方法创建各种简单类型的 Artists

```
import matplotlib.pyplot as plt
X1 = range(0, 50)
Y1 = [num**2 for num in X1] # y = x^2
X2 = [0, 1]
Y2 = [0, 1]    # y = x
Fig = plt.figure(figsize=(8,4))                    # Create a `figure' instance
Ax = Fig.add_subplot(111)                  # Create a `axes' instance in the figure
Ax.plot(X1, Y1, X2, Y2)                    # Create a Line2D instance in the axes
Fig.show()
Fig.savefig("test.pdf")
```

参考：

《Python科学计算》(Numpy视频) matplotlib-绘制精美的图表(快速绘图)(面向对象绘图)（深入浅出适合系统学习）

什么是 Matplotlib （主要讲面向对象绘图，对于新手可能有点乱）

Matplotlib.pylab 快速绘图

matplotlib 还提供了一个名为 pylab 的模块，其中包括了许多 NumPy 和 pyplot 模块中常用的函数，方便用户快速进行计算和绘图，十分适合在 IPython 交互式环境中使用。这里使用下面的方式载入 pylab 模块：

```
>>> import pylab as pl
```

1 安装 numpy 和 matplotlib

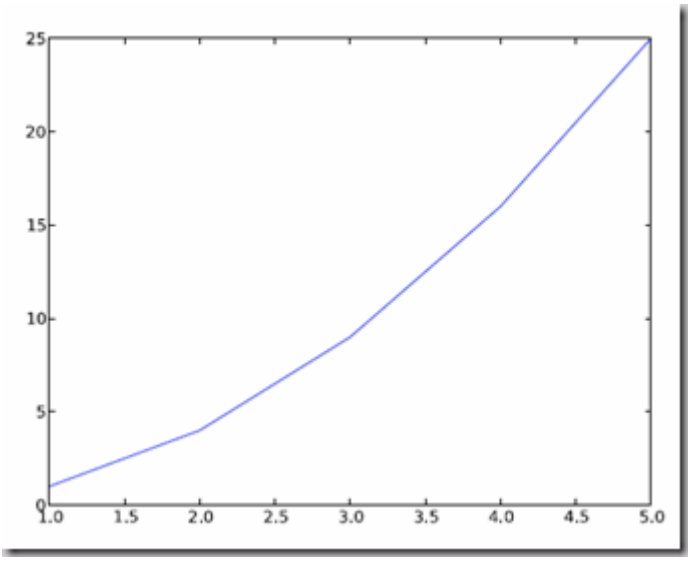
```
>>> import numpy
>>> numpy.__version__
>>> import matplotlib
>>> matplotlib.__version__
```

2 两种常用图类型：Line and scatter plots(使用 plot()命令), histogram(使用 hist()命令)

2.1 折线图&散点图 Line and scatter plots

2.1.1 折线图 Line plots(关联一组 x 和 y 值的直线)

```
import numpy as np
import pylab as pl
x = [1, 2, 3, 4, 5]# Make an array of x values
y = [1, 4, 9, 16, 25]# Make an array of y values for each x value
pl.plot(x, y)# use pylab to plot x and y
pl.show()# show the plot on the screen
```



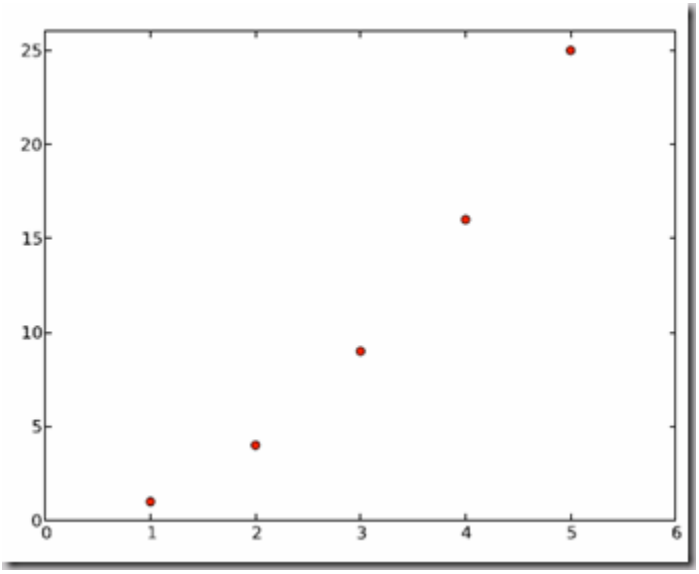
2.1.2 散点图 Scatter plots

把 pl.plot(x, y)改成 pl.plot(x, y, 'o')即可，下图的蓝色版本

2.2 美化 Making things look pretty

2.2.1 线条颜色 Changing the line color

红色：把 pl.plot(x, y, 'o')改成 pl.plot(x, y, 'or')



character	color
b	blue
g	green
r	red
c	cyan
m	magenta
y	yellow
k	black
w	white

2.2.2 线条样式 Changing the line style

虚线:plot(x,y, '--')

linestyle	description
'-'	solid
'--'	dashed
'-.'	dash_dot
':'	dotted
'None'	draw nothing
' '	draw nothing
''	draw nothing

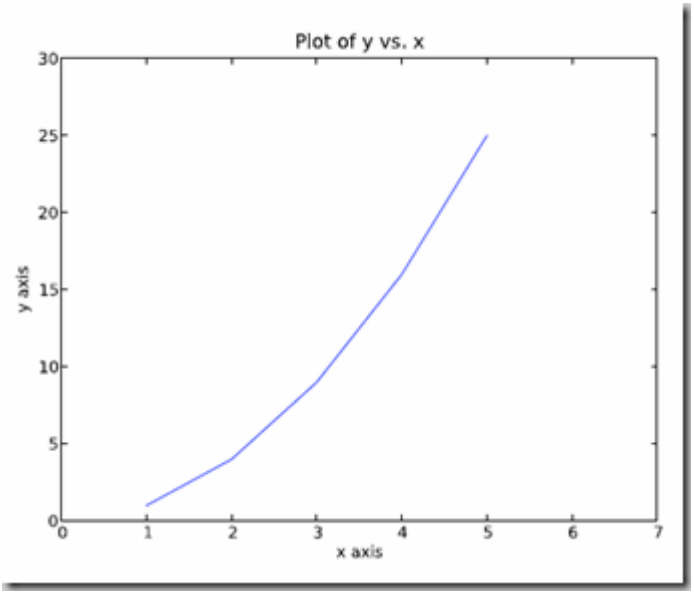
2.2.3 marker 样式 Changing the marker style

蓝色星型 markers：plot(x,y, 'b*')

's'	square marker
'p'	pentagon marker
'*'	star marker
'h'	hexagon1 marker
'H'	hexagon2 marker
'+'	plus marker
'x'	x marker
'D'	diamond marker
'd'	thin diamond marker

2.2.4 图和轴标题以及轴坐标限度 Plot and axis titles and limits

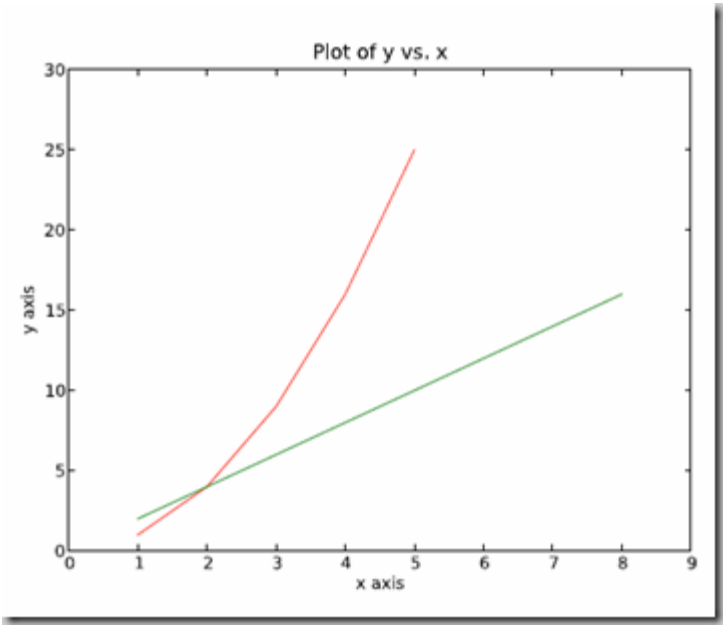
```
import numpy as np
import pylab as pl
x = [1, 2, 3, 4, 5]# Make an array of x values
y = [1, 4, 9, 16, 25]# Make an array of y values for each x value
pl.plot(x, y)# use pylab to plot x and y
pl.title('Plot of y vs. x')# give plot a title
pl.xlabel('x axis')# make axis labels
pl.ylabel('y axis')
pl.xlim(0.0, 7.0)# set axis limits
pl.ylim(0.0, 30.)
pl.show()# show the plot on the screen
```



2.2.5 在一个坐标系上绘制多个图 Plotting more than one plot on the same set of axes

做法是很直接的，依次作图即可：

```
import numpy as np
import pylab as pl
x1 = [1, 2, 3, 4, 5]# Make x, y arrays for each graph
y1 = [1, 4, 9, 16, 25]
x2 = [1, 2, 4, 6, 8]
y2 = [2, 4, 8, 12, 16]
pl.plot(x1, y1, 'r')# use pylab to plot x and y
pl.plot(x2, y2, 'g')
pl.title('Plot of y vs. x')# give plot a title
pl.xlabel('x axis')# make axis labels
pl.ylabel('y axis')
pl.xlim(0.0, 9.0)# set axis limits
pl.ylim(0.0, 30.)
pl.show()# show the plot on the screen
```



2.2.6 图例 Figure legends

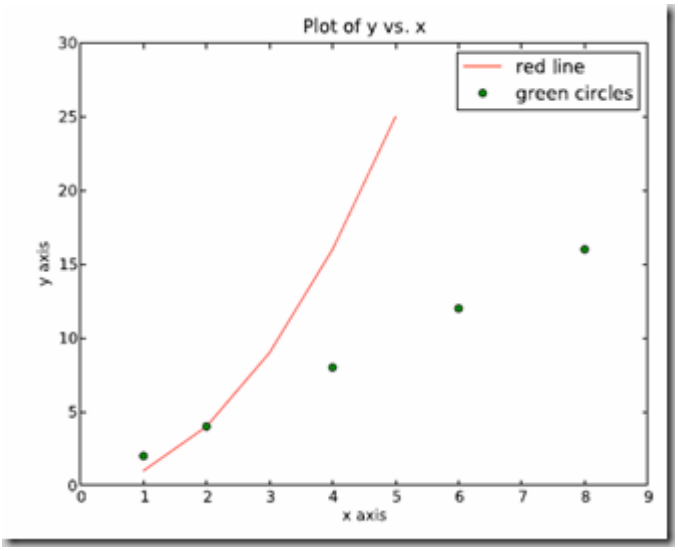
pl.legend((plot1, plot2), ('label1, label2'), 'best', numpoints=1)

其中第三个参数表示图例放置的位置:'best''upper right', 'upper left', 'center', 'lower left', 'lower right'.

如果在当前 figure 里 plot 的时候已经指定了 label，如 plt.plot(x,z,label="$cos(x^2)$")，直接调用 plt.legend()就可以了哦。

```
import numpy as np
import pylab as pl
x1 = [1, 2, 3, 4, 5]# Make x, y arrays for each graph
y1 = [1, 4, 9, 16, 25]
x2 = [1, 2, 4, 6, 8]
```

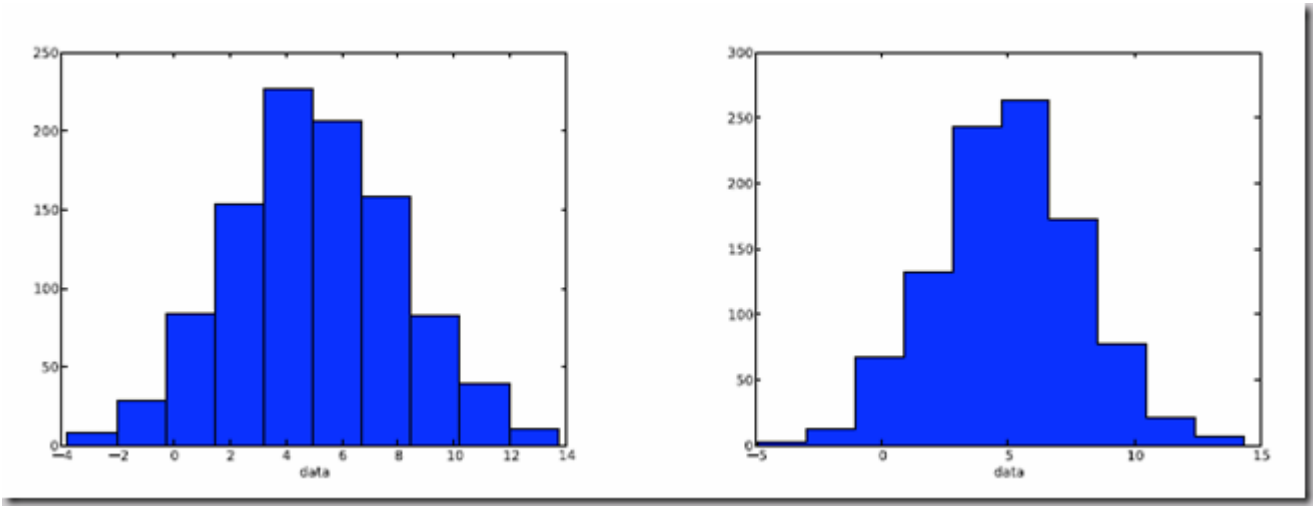
```
y2 = [2, 4, 8, 12, 16]
plot1 = pl.plot(x1, y1, 'r')# use pylab to plot x and y : Give your plots names
plot2 = pl.plot(x2, y2, 'go')
pl.title('Plot of y vs. x')# give plot a title
pl.xlabel('x axis')# make axis labels
pl.ylabel('y axis')
pl.xlim(0.0, 9.0)# set axis limits
pl.ylim(0.0, 30.)
pl.legend([plot1, plot2], ('red line', 'green circles'), 'best', numpoints=1)# make legend
pl.show()# show the plot on the screen
```



2.3 直方图 Histograms

```
import numpy as np
import pylab as pl
# make an array of random numbers with a gaussian distribution with
# mean = 5.0
# rms = 3.0
# number of points = 1000
data = np.random.normal(5.0, 3.0, 1000)
# make a histogram of the data array
pl.hist(data)
# make plot labels
pl.xlabel('data')
pl.show()
```

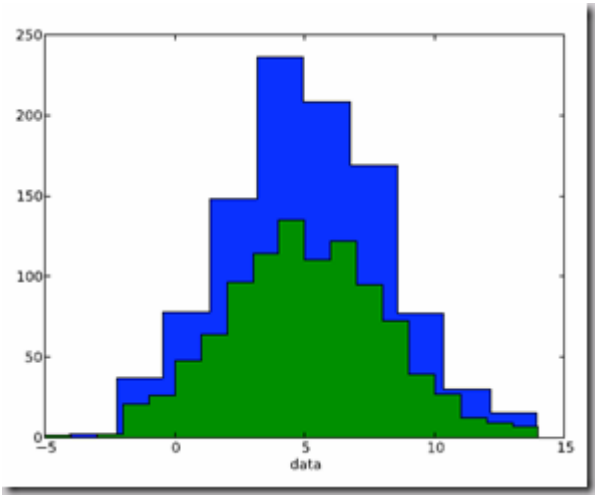
如果不想要黑色轮廓可以改为 `pl.hist(data, histtype='stepfilled')`



2.3.1 自定义直方图 bin 宽度 Setting the width of the histogram bins manually

增加这两行

```
bins = np.arange(-5., 16., 1.) #浮点数版本的 range
pl.hist(data, bins, histtype='stepfilled')
```



3 同一画板上绘制多幅子图 Plotting more than one axis per canvas

如果需要同时绘制多幅图表的话，可以是给 figure 传递一个整数参数指定图标的序号，如果所指定
序号的绘图对象已经存在的话，将不创建新的对象，而只是让它成为当前绘图对象。

```
fig1 = pl.figure(1)
pl.subplot(211)
```

subplot(211)把绘图区域等分为 2 行*1 列共两个区域，然后在区域 1(上区域)中创建一个轴对

象. pl.subplot(212)在区域 2(下区域)创建一个轴对象。

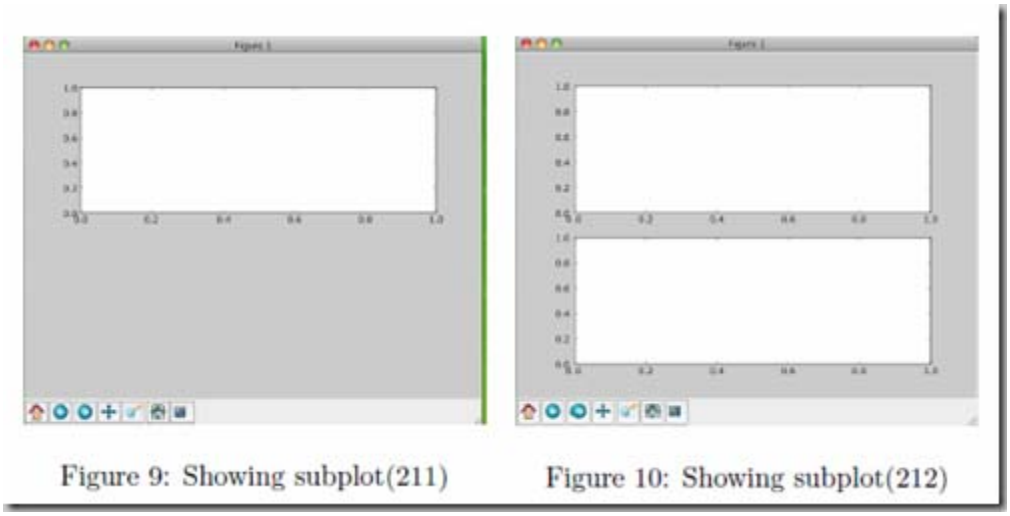


Figure 9: Showing subplot(211)　　Figure 10: Showing subplot(212)

You can play around with plotting a variety of layouts. For example, Fig. 11 is created using the following commands:

```
f1 = pl.figure(1)
pl.subplot(221)
pl.subplot(222)
pl.subplot(212)
```

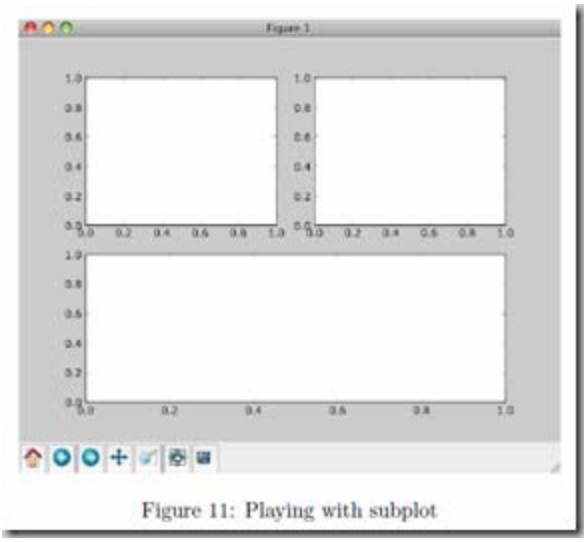


Figure 11: Playing with subplot

当绘图对象中有多个轴的时候，可以通过工具栏中的 Configure Subplots 按钮，交互式地调节轴之间的间距和轴与边框之间的距离。如果希望在程序中调节的话，可以调用 subplots_adjust 函数，它有 left, right, bottom, top, wspace, hspace 等几个关键字参数，这些参数的值都是 0 到 1 之间的小数，它们是以绘图区域的宽高为 1 进行正规化之后的坐标或者长度。

```
pl.subplots_adjust(left=0.08, right=0.95, wspace=0.25, hspace=0.45)
```

4 绘制文件中的数据 Plotting data contained in files

4.1 从 Ascii 文件中读取数据 Reading data from ascii files

读取文件的方法很多，这里只介绍一种简单的方法，更多的可以参考官方文档和NumPy快速处理数据(文件存取)。

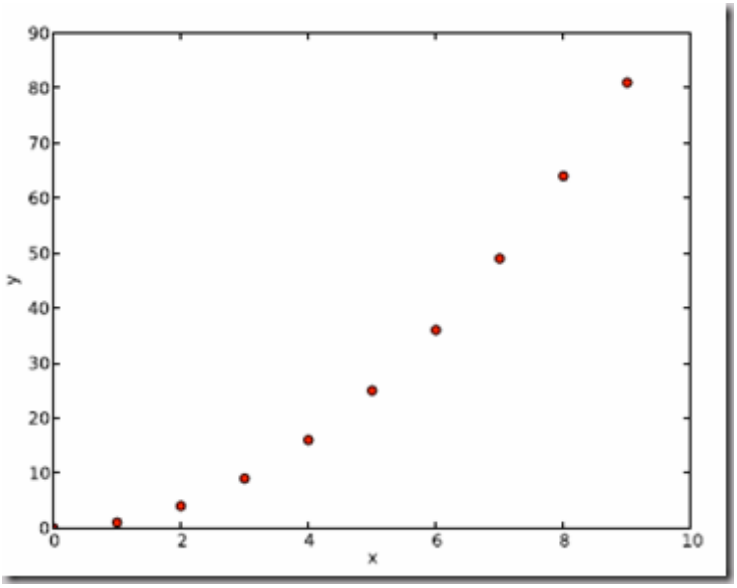
numpy 的 loadtxt 方法可以直接读取如下文本数据到 numpy 二维数组

****

```
# fakedata.txt
0 0
1 1
2 4
3 9
4 16
5 25
6 36
7 49
8 64
9 81
0 0
1 1
2 4
3 9
4 16
5 25
6 36
7 49
8 64
9 81
```

****

```
import numpy as np
import pylab as pl
# Use numpy to load the data contained in the file
# 'fakedata.txt' into a 2-D array called data
data = np.loadtxt('fakedata.txt')
# plot the first column as x, and second column as y
pl.plot(data[:,0], data[:,1], 'ro')
pl.xlabel('x')
pl.ylabel('y')
pl.xlim(0.0, 10.)
pl.show()
```



4.2 写入数据到文件 Writing data to a text file

写文件的方法也很多，这里只介绍一种可用的写入文本文件的方法，更多的可以参考官方文档。

```
import numpy as np
# Let's make 2 arrays (x, y) which we will write to a file
# x is an array containing numbers 0 to 10, with intervals of 1
x = np.arange(0.0, 10., 1.)
# y is an array containing the values in x, squared
y = x*x
print 'x = ', x
print 'y = ', y
# Now open a file to write the data to
# 'w' means open for 'writing'
file = open('testdata.txt', 'w')
# loop over each line you want to write to file
for i in range(len(x)):
    # make a string for each line you want to write
    # '\t' means 'tab'
    # '\n' means 'newline'
    # 'str()' means you are converting the quantity in brackets to a string type
    txt = str(x[i]) + '\t' + str(y[i]) + ' \n'
    # write the txt to the file
    file.write(txt)
# Close your file
file.close()
```

这部分是翻译自：Python Plotting Beginners Guide

对 LaTeX 数学公式的支持

Matlplotlib 对 LaTeX 有一定的支持，如果记得使用 raw 字符串语法会很自然：

```
xlabel(r"$\frac{x^2}{y^4}$")
```

在 matplotlib 里面，可以使用 LaTex 的命令来编辑公式，只需要在字符串前面加一个“r”即可

Here is a simple example:

```
# plain text
plt.title('alpha > beta')
```

produces “alpha > beta”.

Whereas this:

```
# math text
plt.title(r'$\alpha > \beta$')
```

produces "$\alpha > \beta$".

这里给大家看一个简单的例子。

```
import matplotlib.pyplot as plt
x = arange(1,1000,1)
r = -2
c = 5
y = [5*(a**r) for a in x]


fig = plt.figure()
ax = fig.add_subplot(111)
ax.loglog(x,y,label = r"$y = \frac{1}{2\sigma_1^2},
c=5,\sigma_1=-2$")
ax.legend()
ax.set_xlabel(r"x")
ax.set_ylabel(r"y")
```

程序执行结果如图 3 所示，这实际上是一个 power-law 的例子，有兴趣的朋友可以继续 google 之。

再看一个《用 Python 做科学计算》中的简单例子，下面的两行程序通过调用 plot 函数在当前的绘图对象中进行绘图：

```
plt.plot(x,y,label="$sin(x)$",color="red",linewidth=2)

plt.plot(x,z,"b--",label="$cos(x^2)$")
```

plot 函数的调用方式很灵活，第一句将 x,y 数组传递给 plot 之后，用关键字参数指定各种属性：

- **label**：给所绘制的曲线一个名字，此名字在图示(legend)中显示。只要在字符串前后添加"**$**"符号，matplotlib 就会使用其**内嵌的 latex 引擎**绘制的数学公式。
- **color**：指定曲线的颜色
- **linewidth**：指定曲线的宽度

详细的可以参考 matplotlib 官方教程:

Writing mathematical expressions

- Subscripts and superscripts
- Fractions, binomials and stacked numbers
- Radicals
- Fonts
    - Custom fonts
- Accents
- Symbols
- Example

Text rendering With LaTeX

- usetex with unicode
- Postscript options
- Possible hangups
- Troubleshooting

有几个问题：

- matplotlib.rcParams 属性字典
- 想要它正常工作，在 matplotlibrc 配置文件中需要设置 text.markup = "tex"。
- 如果你希望图表中所有的文字（包括坐标轴刻度标记）都是 LaTeX'd,需要在 matplotlibrc 中设置 text.usetex = True。如果你使用 LaTeX 撰写论文，那么这一点对于使图表和论文中其余部分保持一致是很有用的。
- 在 matplotlib 中使用中文字符串时记住要用 unicode 格式，例如：u''测试中文显示''

matplotlib使用小结

LaTeX科技排版

参考文献自动搜集管理完美攻略（图文版）:Latex+Lyx+Zotero

对数坐标轴

在实际中，我们可能经常会用到对数坐标轴，这时可以用下面的三个函数来实现

ax.semilogx(x,y) #x 轴为对数坐标轴

ax.semilogy(x,y) #y 轴为对数坐标轴

ax.loglog(x,y) #双对数坐标轴

学习资源

Gnuplot 的介绍

- Gnuplot简介
- IBM:gnuplot 让您的数据可视化,Linux 上的数据可视化工具
- 利用Python绘制论文图片: Gnuplot,pylab

官方英文资料：

- matplotlib下载及API手册地址
- Screenshots:example figures
- Gallery:Click on any image to see full size image and source code
- matplotlib所使用的数学库numpy下载及API手册

IBM:基于 Python Matplotlib 模块的高质量图形输出(2005 年的文章有点旧)

matplotlib技巧集(绘制不连续函数的不连续点；参数曲线上绘制方向箭头；修改缺省刻度数目；Y轴不同区间使用不同颜色填充的曲线区域。)

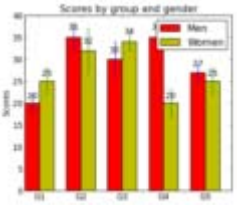
Python:使用matp绘制不连续函数的不连续点；参数曲线上绘制方向箭头；修改缺省刻度数目；Y轴不同区间使用不同颜色填充的曲线区域。lotlib绘制图表

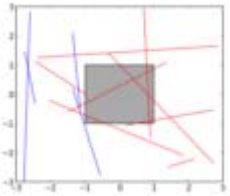
matplotlib输出图象的中文显示问题

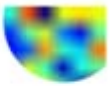
matplotlib图表中图例大小及字体相关问题

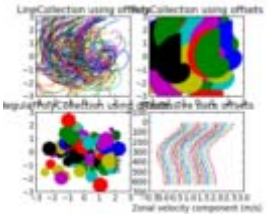
Click on any image to see full size image and source code

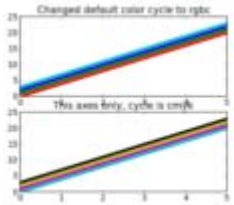


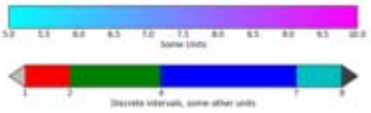


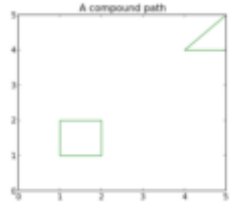


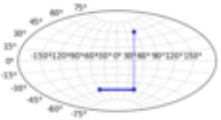


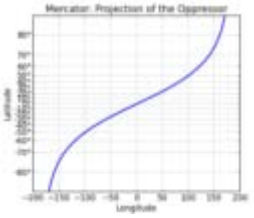


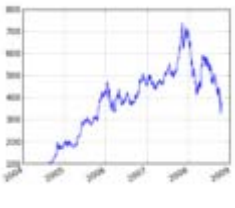


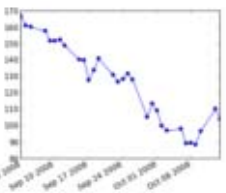


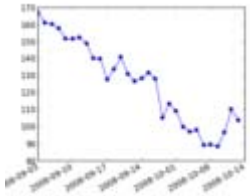


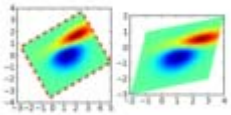


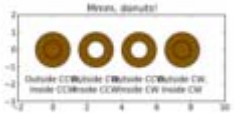


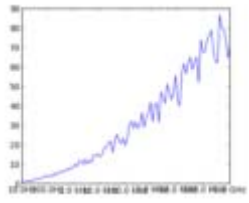


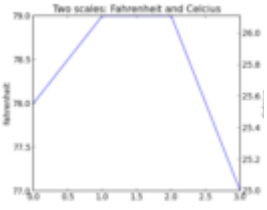


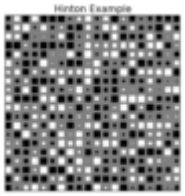


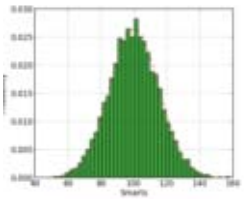


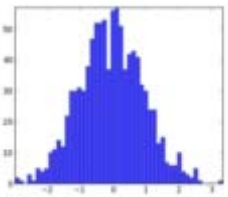


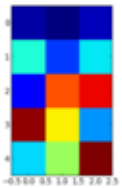


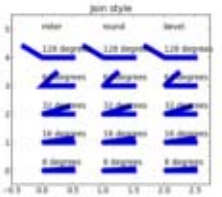


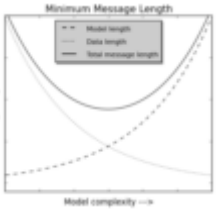


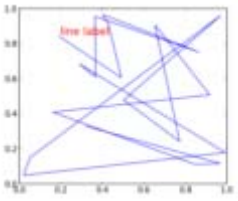








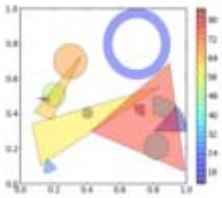


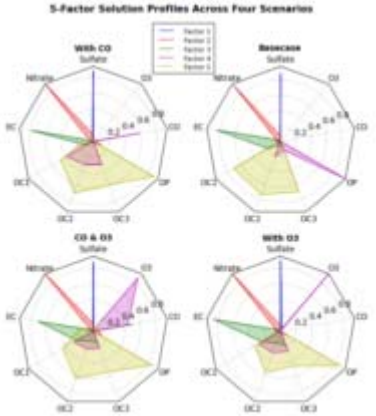


matplotlib

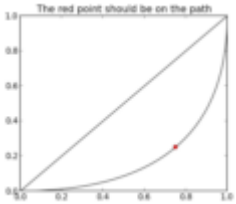
some other

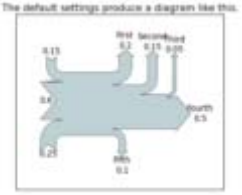
IQ: $\sigma_i = 15$



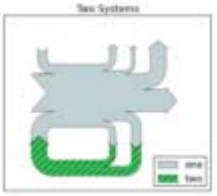








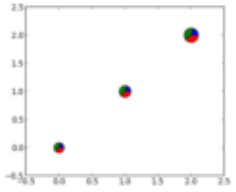


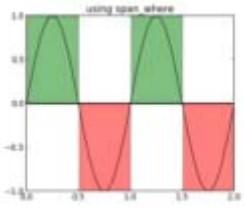


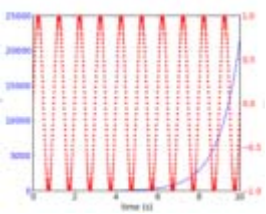


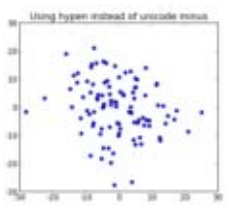


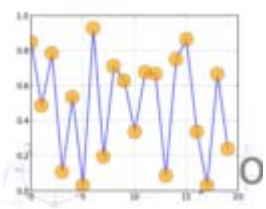




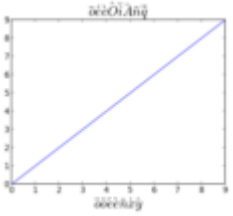


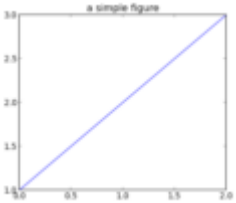


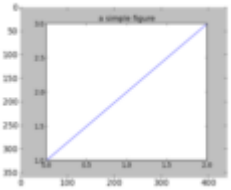




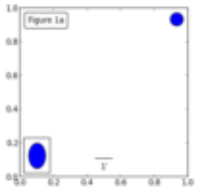

Property of MPL

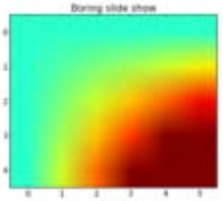


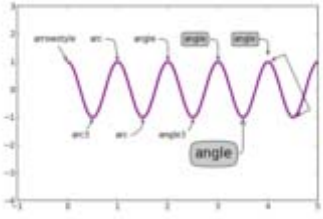

a simple figure

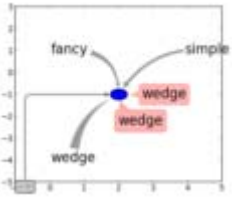

a simple figure

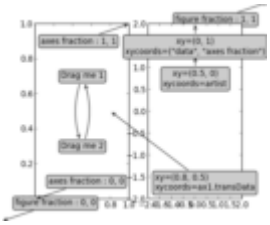

right bottom
middle
center top

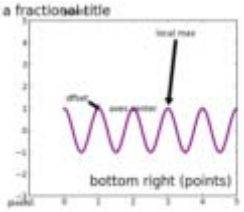

Figure 1x

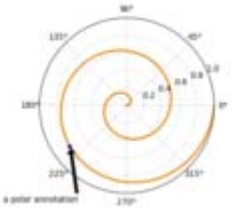

Boring slide show

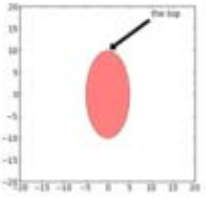

arrowstyle
arc
angle
angle
angle
angle
arc3
arc
angle3

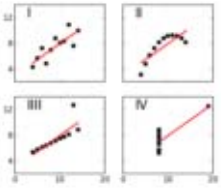

fancy
simple
wedge
wedge
wedge

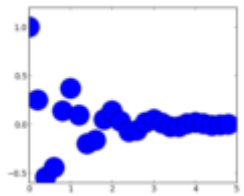

figure fraction : 1, 1
axes fraction : 1, 1
Drag me 1
Drag me 2
axes fraction : 0, 0
figure fraction : 0, 0

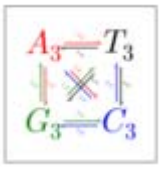

a fractional title
local max
offset
axes center
bottom right (points)

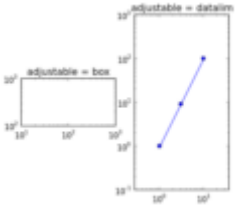

a polar annotation

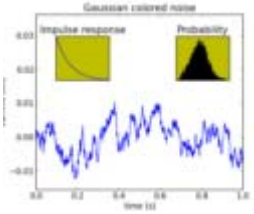

the top


I
II
III
IV

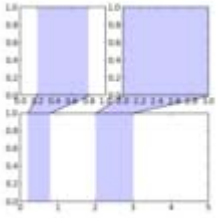


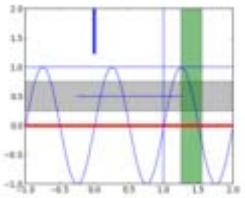

A_3
T_3
G_3
C_3

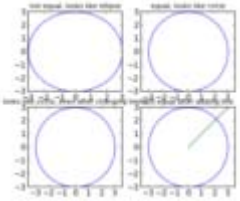

adjustable = datalim
adjustable = box

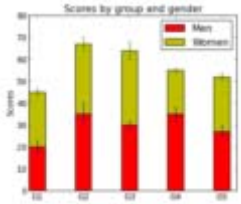

Gaussian colored noise
Impulse response
Probability
time (s)

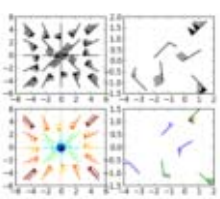


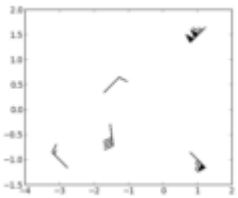


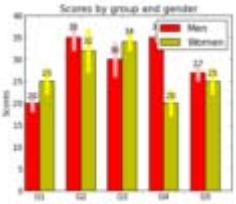


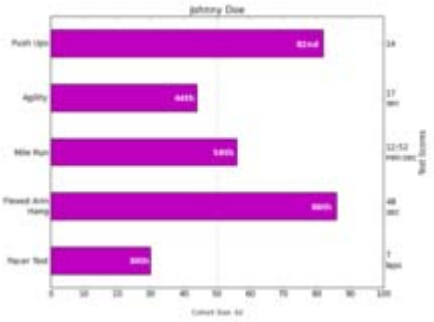



Scores by group and gender
Men
Women
Scores

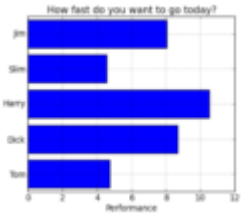


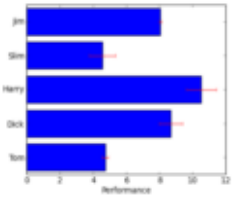


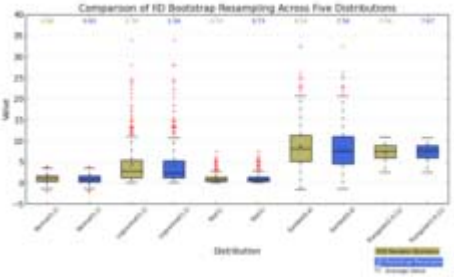

Scores by group and gender
Men
Women
Scores

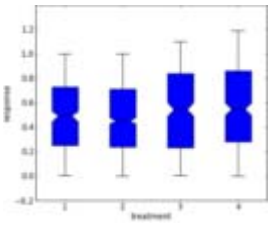


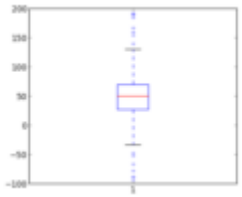


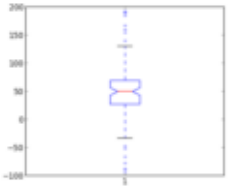


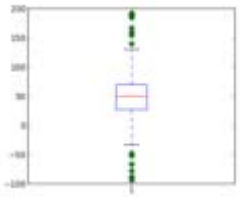


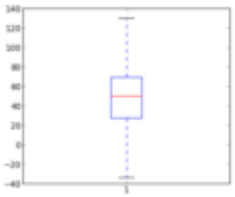


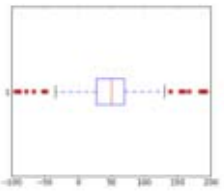


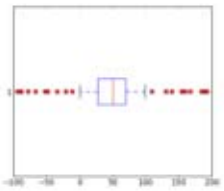


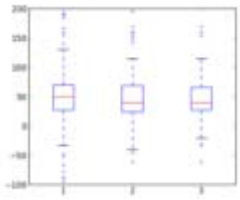




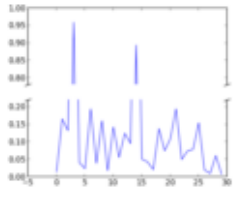


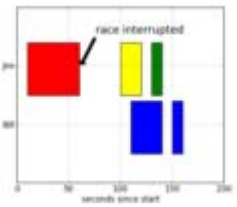


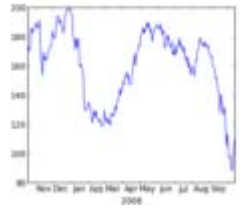


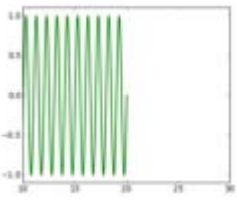


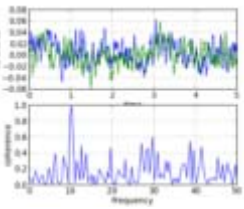
Distance Histograms by Categor

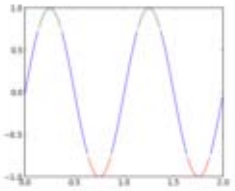


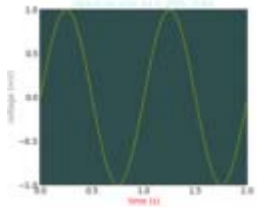


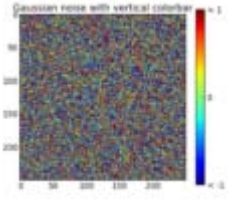


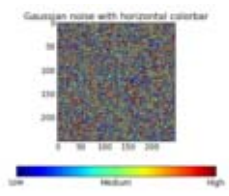


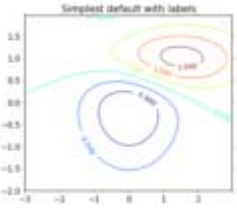


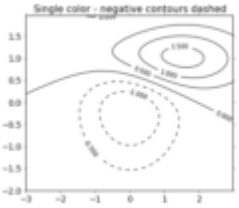


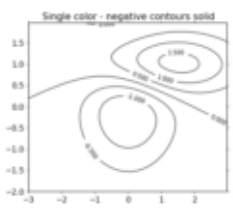


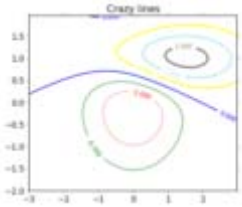


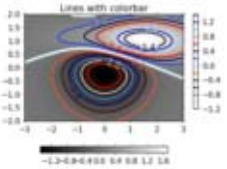

Gaussian noise with horizontal colorbar
Low
Medium
High

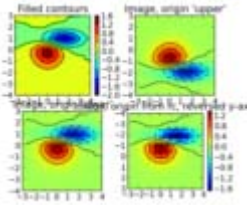

Simplest default with labels

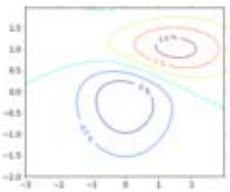

Single color - negative contours dashed

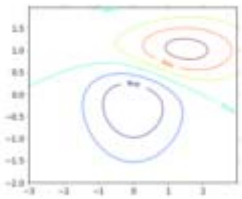

Single color - negative contours solid

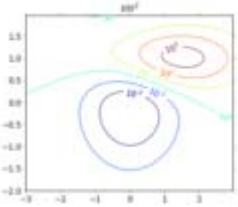

Crazy lines

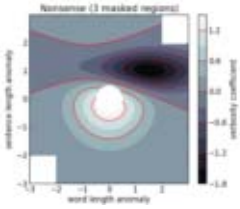

Lines with colorbar

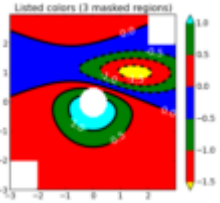

Filled contours
Image, origin 'upper'

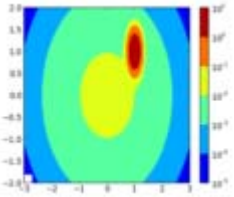


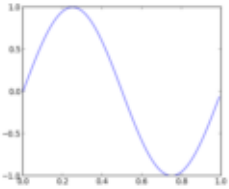

Rez

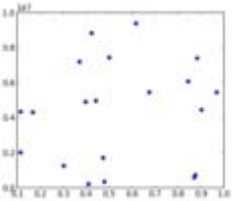

10^2
10^1
10^0

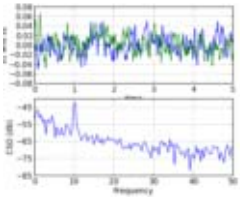

Nonsense (3 masked regions)
sentence length anomaly
word length anomaly
verbosity coefficient

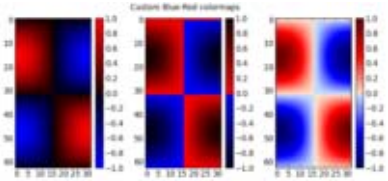

Listed colors (3 masked regions)

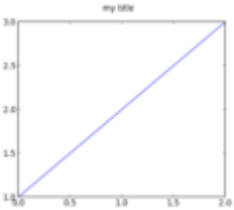


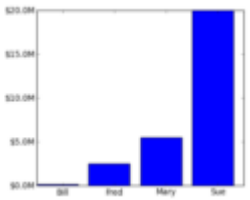


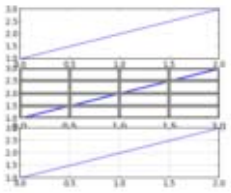


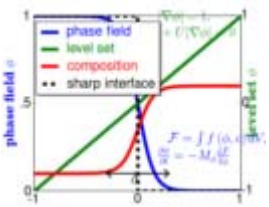

Frequency

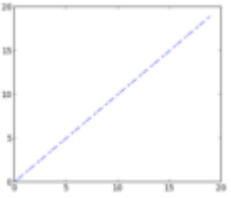

Custom Blue-Red colormaps

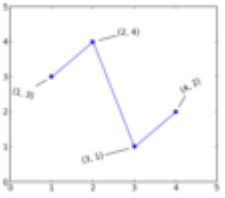

my title

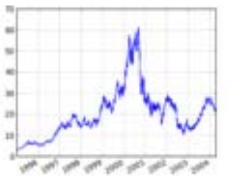

$20.0M
$15.0M
$10.0M
$5.0M
$0.0M
Bill
Fred
Mary
Sue

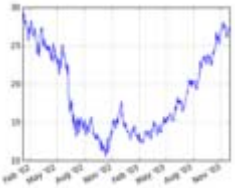


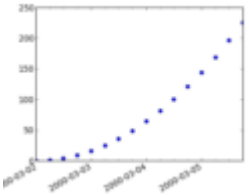

phase field
level set
composition
sharp interface
phase field ϕ
level set ψ

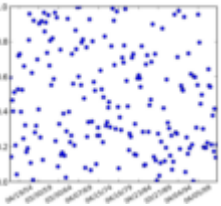


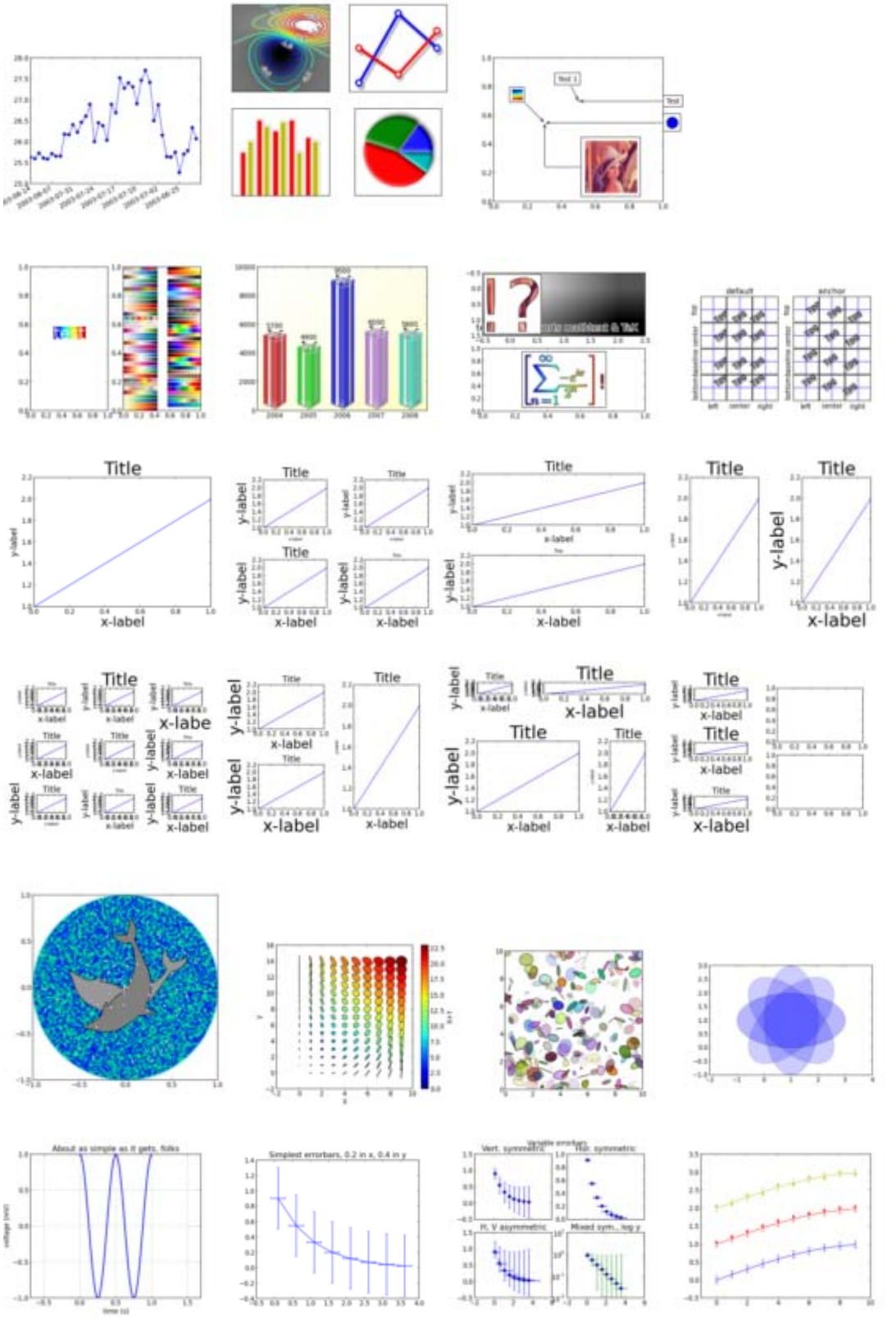

(2, 4)
(0, 2)
(4, 2)
(3, 1)

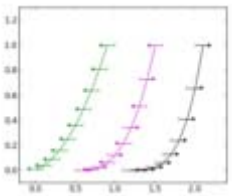

1998
1999
2000
2001
2002
2003
2004

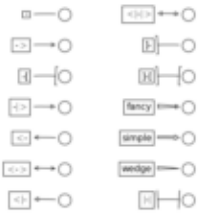

Feb '02
May '02
Aug '02
Nov '02
Feb '03
May '03
Aug '03
Nov '03

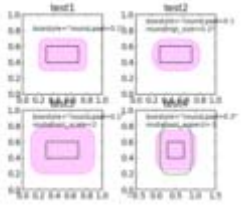

2000-03-02
2000-03-03
2000-03-04
2000-03-05




Title
x-label
y-label
Test 1
Test
default
anchor
left
center
right
About as simple as it gets, folks
Simplest errorbars, 0.2 in x, 0.4 in y
Vert. symmetric
Hor. symmetric
H, V asymmetric
Mixed sym., log y

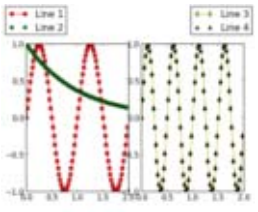


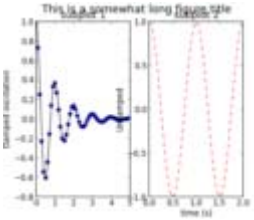

square
sawtooth
roundtooth
rarrow
larrow
round4
round
fancy
simple
wedge

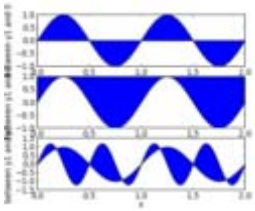

test1
test2

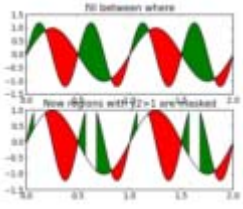

test
test

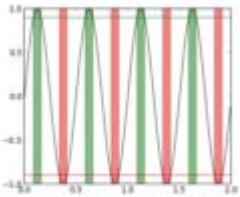


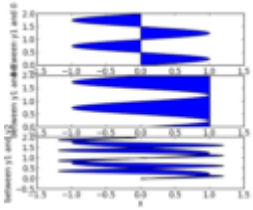

Line 1
Line 2
Line 3
Line 4

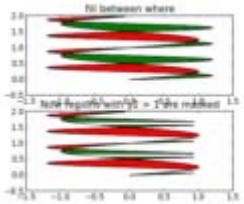

This is a somewhat long figure title

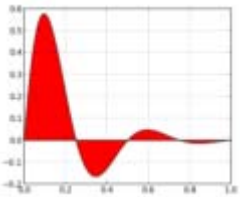


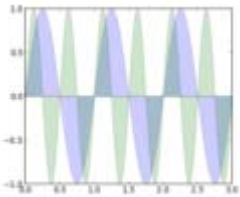

fill between where

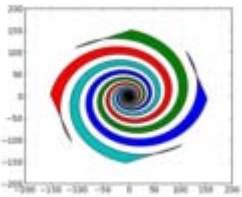


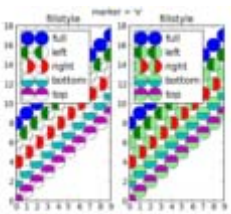


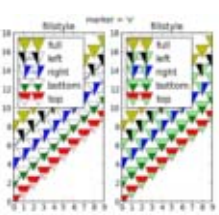

fill between where

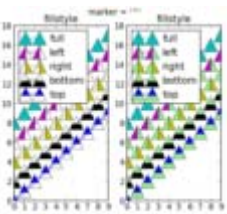


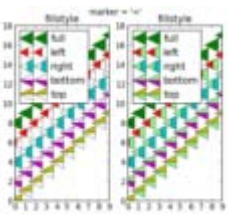


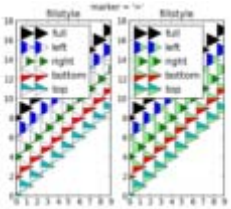


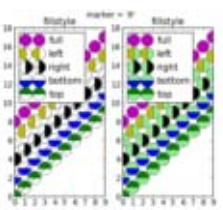

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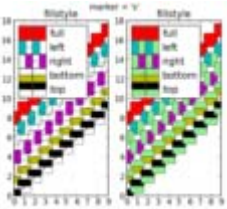

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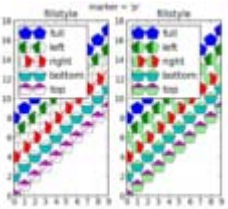

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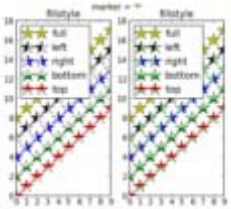

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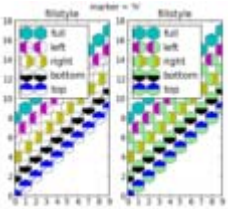

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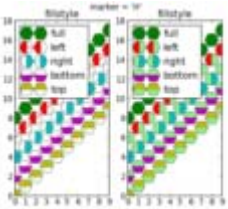

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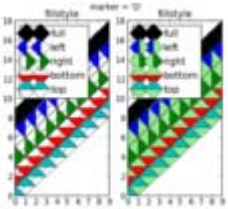

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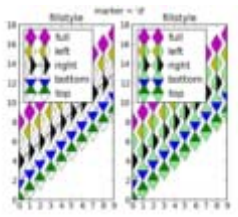

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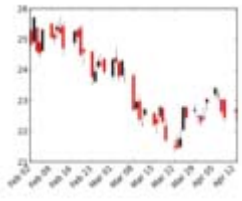

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Bilstyle
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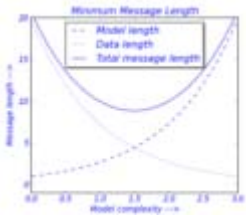

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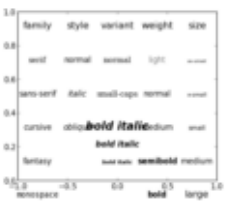

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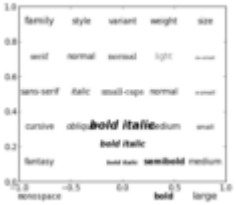

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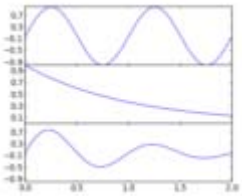


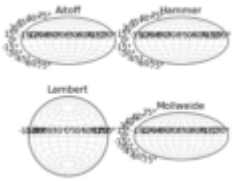

SPY daily

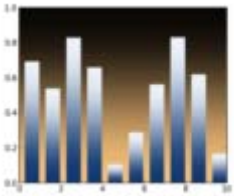

Minimum Message Length
Model length
Data length
Total message length
Message length -->
Model complexity -->

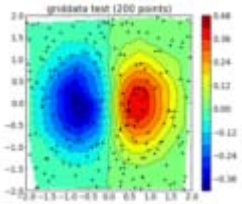

family
style
variant
weight
size
sans-serif
cursive
fantasy
bold italic
bold
large

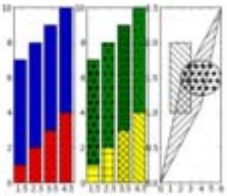

family
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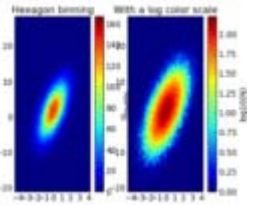


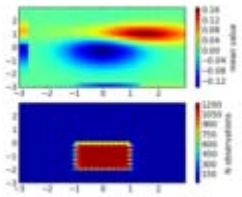

Aitoff
Hammer
Lambert
Mollweide

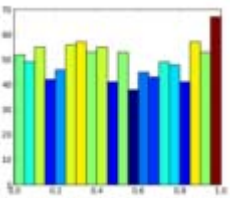


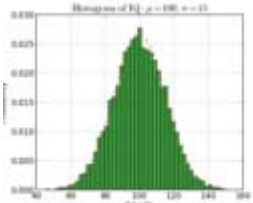

griddata test (200 points)

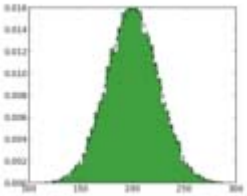


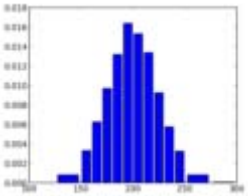

Hexagon binning
With a log color scale

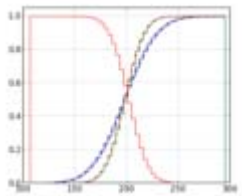

mean value
observations

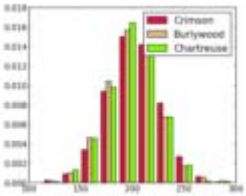


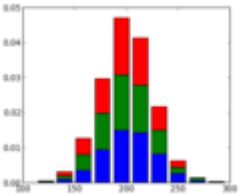


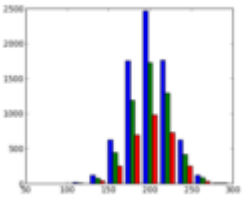


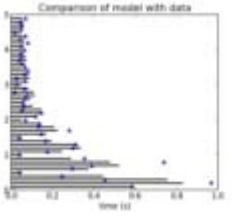


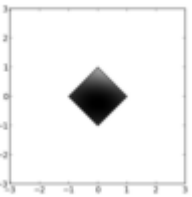


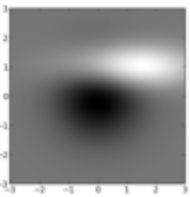

Crimson
Burlywood
Chartreuse

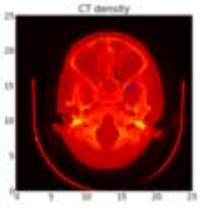


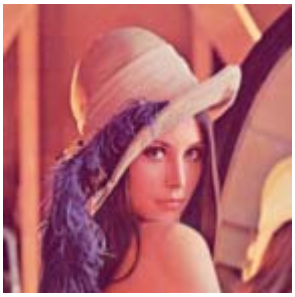


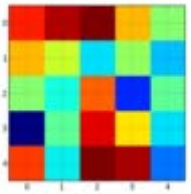

Comparison of model with data

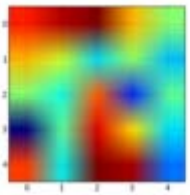


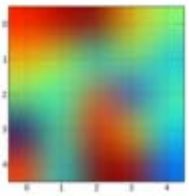


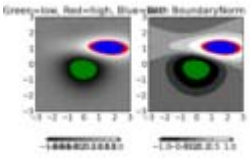

CT density

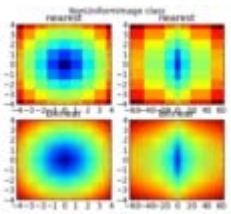


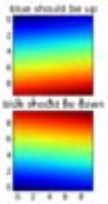


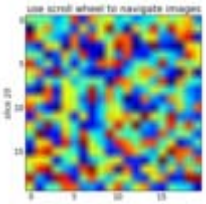


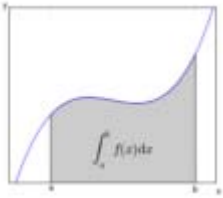


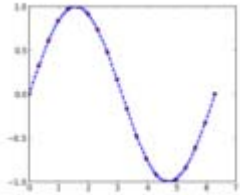


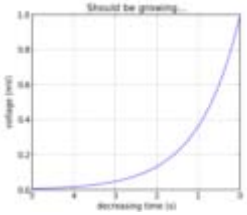


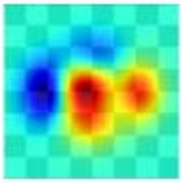

blue should be up
blue should be down

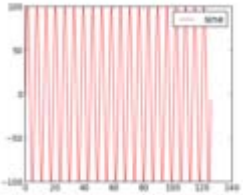

use scroll wheel to navigate images

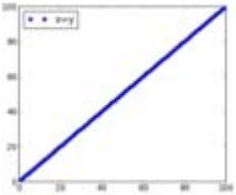

$\int_a^b f(x)dx$

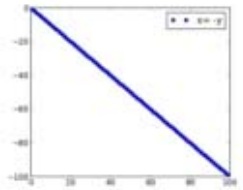


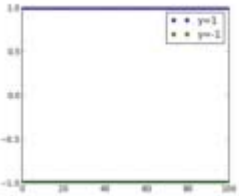

Should be growing...
voltage (mV)
decreasing time (s)

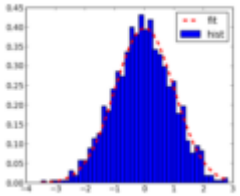


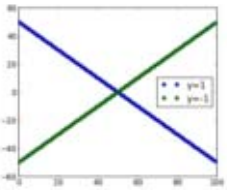

sine

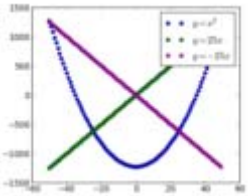

y=x

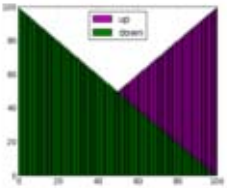

y=-x

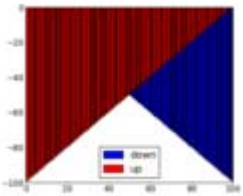

y=1
y=-1

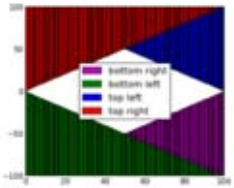

fit
hist

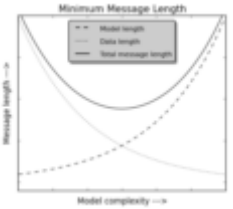

y=1
y=-1

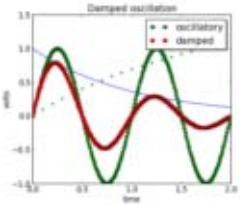


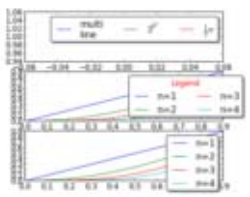

up
down

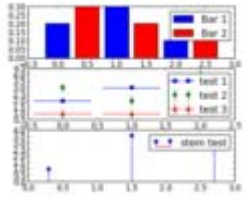

down
up

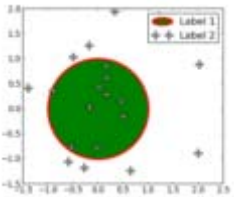

bottom right
bottom left
top left
top right

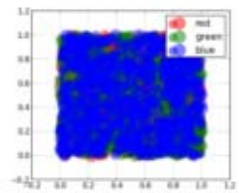

Minimum Message Length
Model length
Data length
Total message length
Message length --->
Model complexity --->

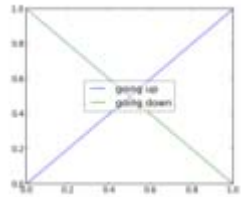

Damped oscillation
oscillatory
damped
time

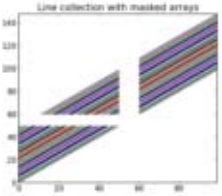

multi
line
Legend

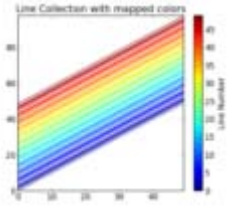

Bar 1
Bar 2
test 1
test 2
test 3
stem test

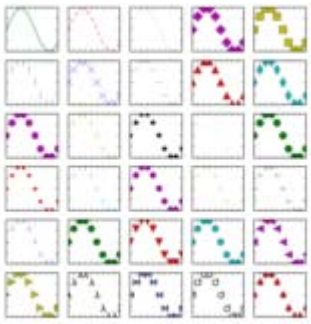

Label 1
Label 2

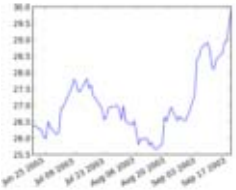

red
green
blue

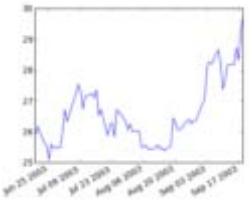

going up
going down

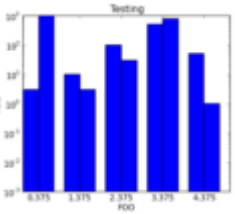

Line collection with masked arrays

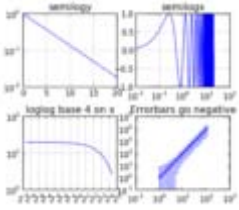

Line Collection with mapped colors
Line number

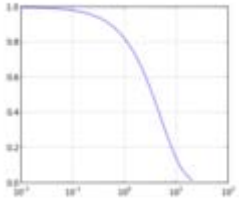


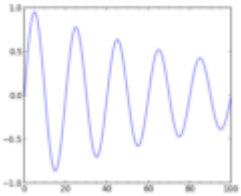


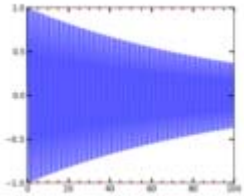



Testing
FOO
FOO

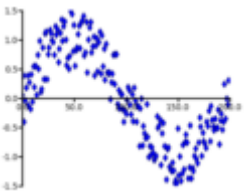

semilogy
semilogx
loglog base 4 on x
Errorbars go negative

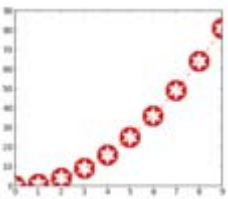


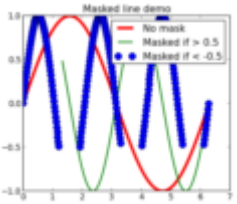


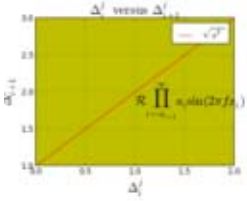


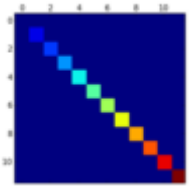

matplotlib

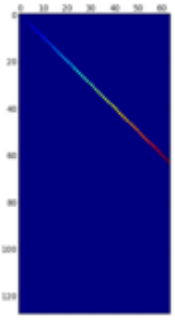


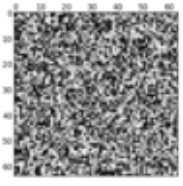


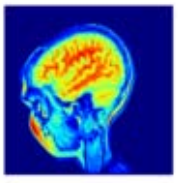

Masked line demo
No mask
Masked if > 0.5
Masked if < -0.5

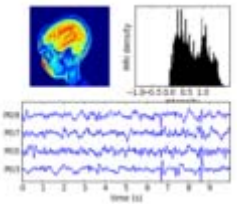


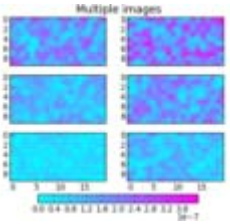


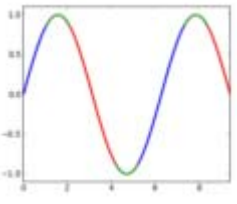


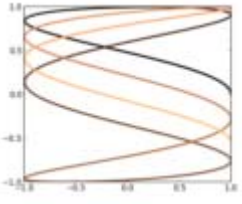


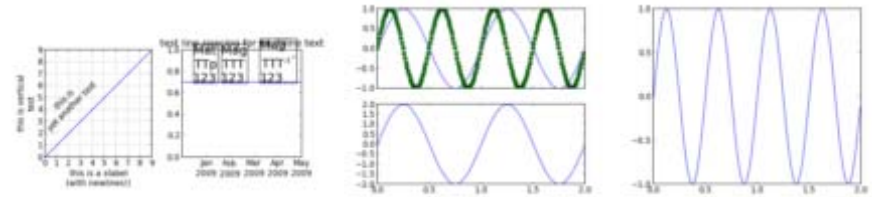


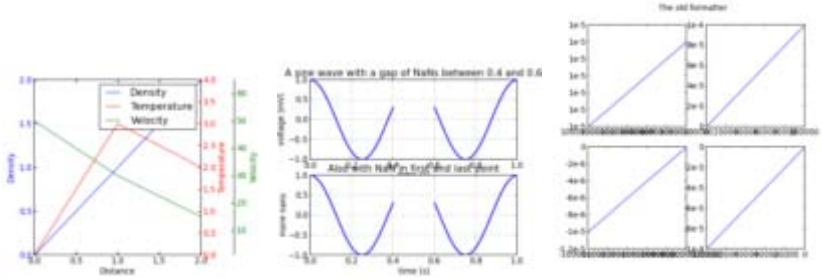



Multiple images

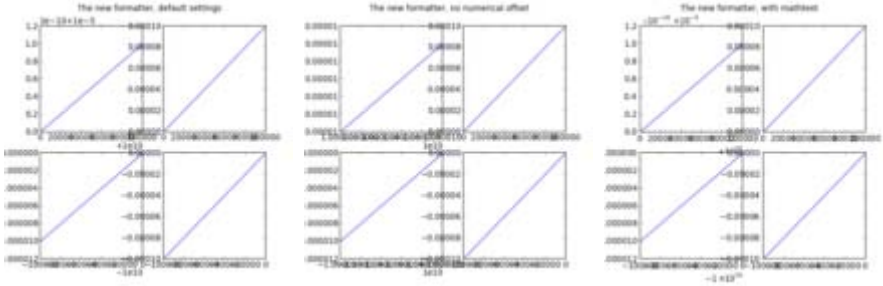


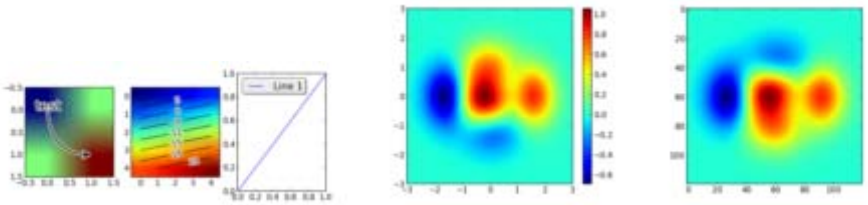


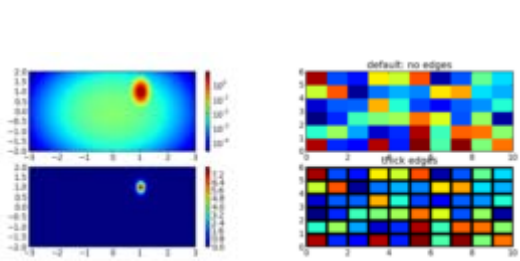


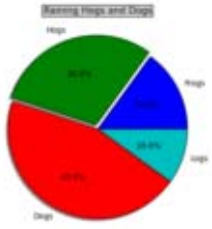

Density
Temperature
Velocity
Distance
A sine wave with a gap of NaNs between 0.4 and 0.6
A cosine wave with a NaN in front and last point
time (s)

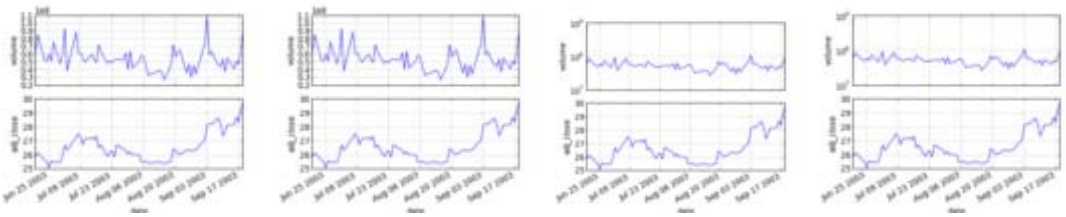
The old formatter

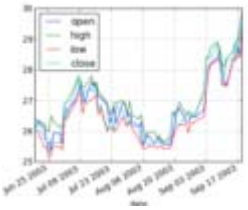

The new formatter, default settings
The new formatter, no numerical offset
The new formatter, with mathtext

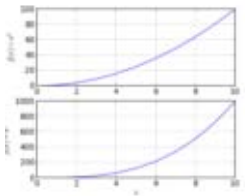

Line 1

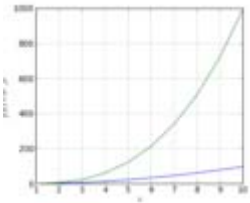

default: no edges
tflick edges

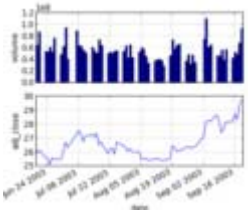

Raining Hogs and Dogs
Hogs
Dogs
Logs

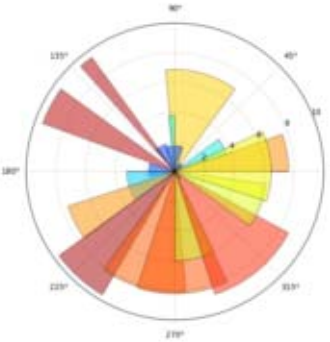

volume
adj_close

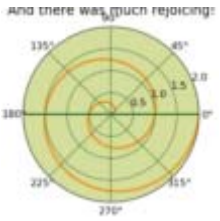

open
high
low
close

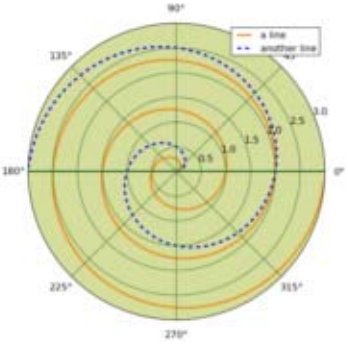


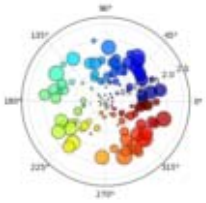


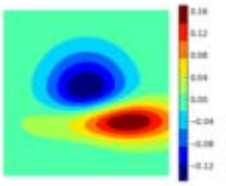

Jet
volume
adj_close

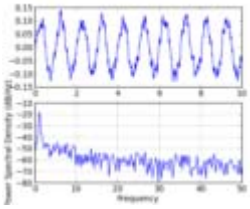


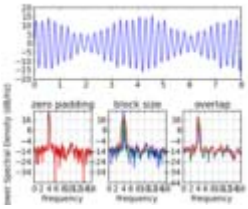

And there was much rejoicing!

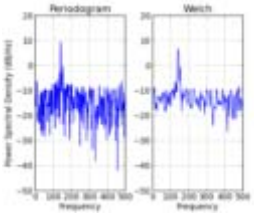

a line
another line

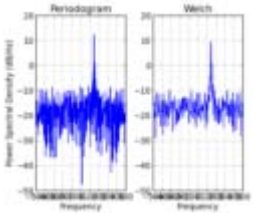


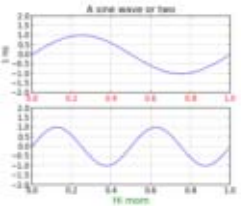


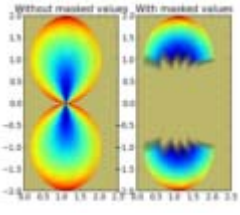

Power Spectral Density (dB/Hz)
Frequency

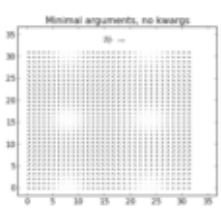

zero padding
block size
overlap
Power Spectral Density (dB/Hz)
Frequency

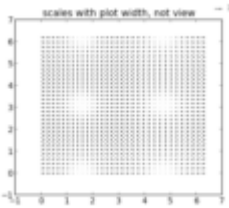

Periodogram
Welch
Power Spectral Density (dB/Hz)
Frequency

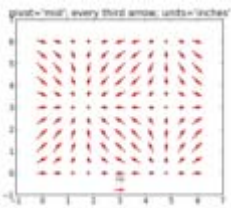

Periodogram
Welch
Power Spectral Density (dB/Hz)
Frequency

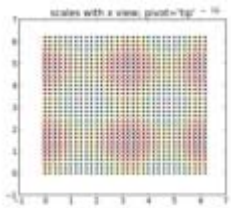

A sine wave or two
Hi mom

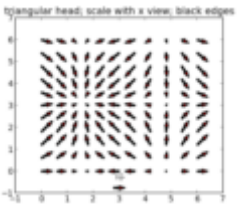

Without masked values
With masked values

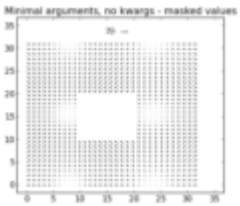

Minimal arguments, no kwargs

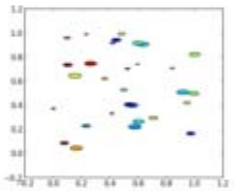

scales with plot width, not view

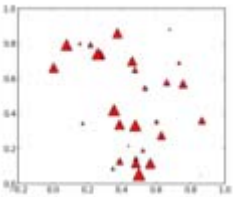

pivot='mid'; every third arrow; units='inches'

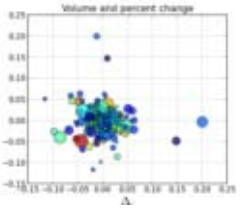

scales with x view; pivot='tip'

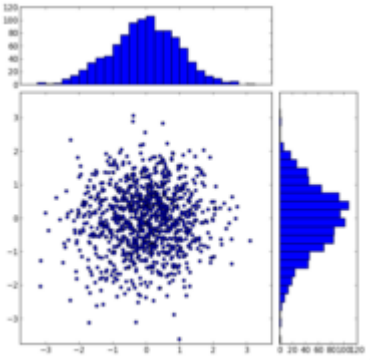

triangular head; scale with x view; black edges

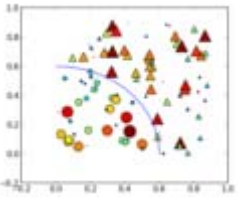

Minimal arguments, no kwargs - masked values

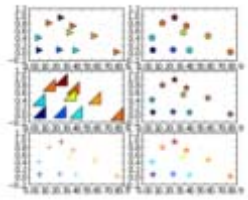


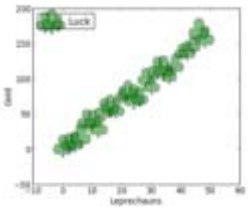


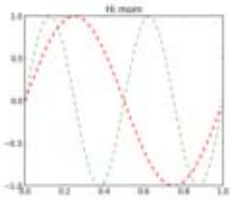

Volume and percent change

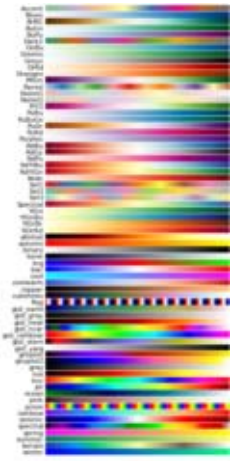


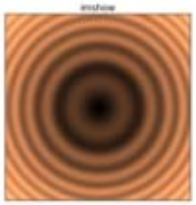




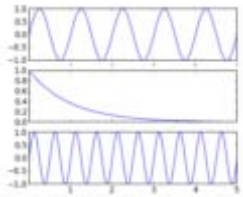

Luck
Gold
Leprechauns

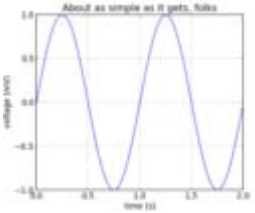

Hi mom

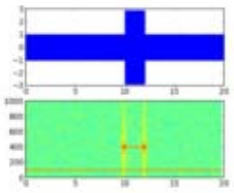


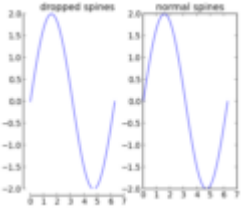

imshow

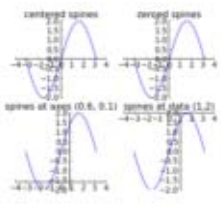

imshow with shading

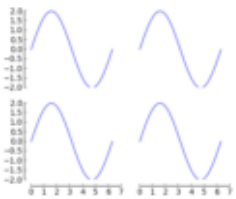


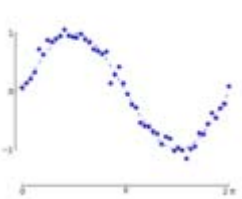

About as simple as it gets, folks
voltage (mV)
time (s)

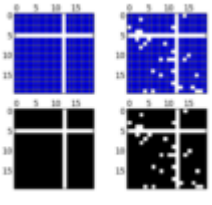


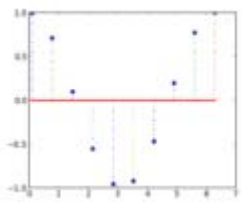

dropped spines
normal spines

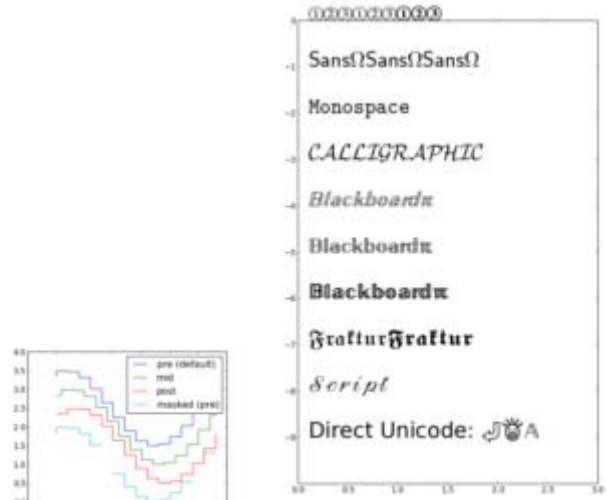

centered spines
zeroed spines
spines at axes (0.6, 0.1)
spines at data (1,2)

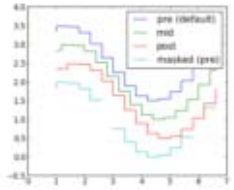


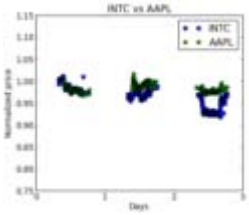


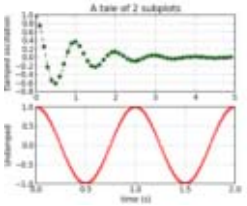


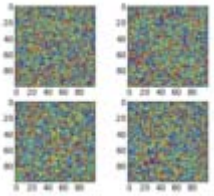



Sans Ω Sans Ω Sans Ω
Monospace
CALLIGRAPHIC
Blackboard 𝔹
Blackboard 𝔹
Blackboard 𝔹
Fraktur Fraktur
Script
Direct Unicode: ♫♛A

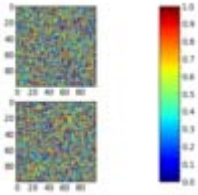

pre (default)
mid
post
masked (pre)

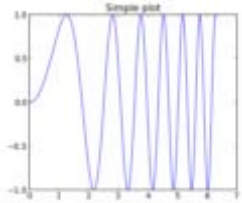

INTC vs AAPL
INTC
AAPL
Days

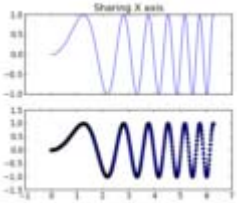

A tale of 2 subplots
time (s)

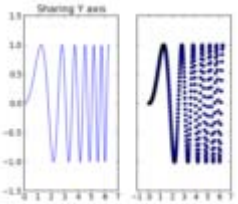


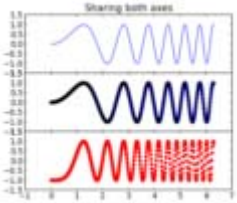

Click on slider to adjust subplot param
left
bottom
right
top
wspace
hspace
Reset

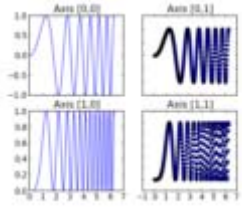


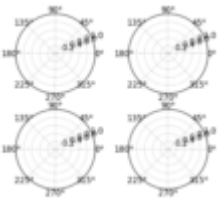

Simple plot

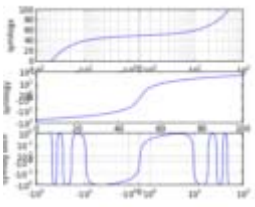

Sharing X axis

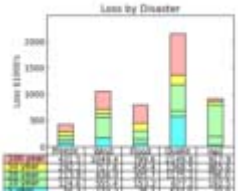

Sharing Y axis


Sharing both axes

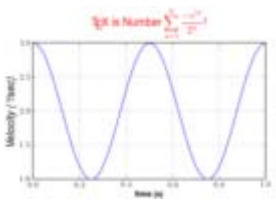

Axis [0,0]
Axis [0,1]
Axis [1,0]
Axis [1,1]

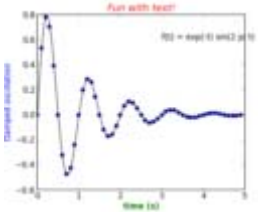


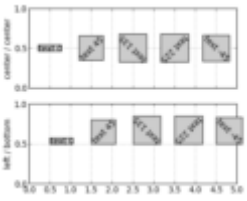


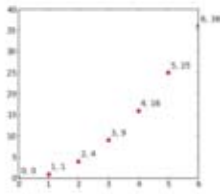

Loss by Disaster

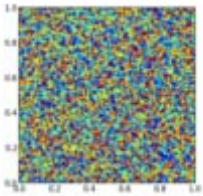


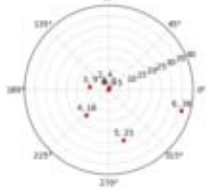


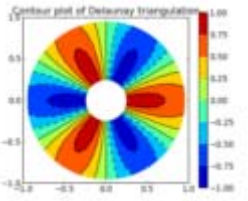

Fun with text!

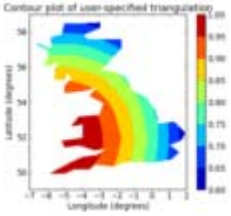

center / center
left / bottom
text not rotated correctly
text rotated correctly
Damped exponential decay
voltage (mV)
time (s)

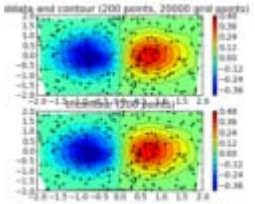


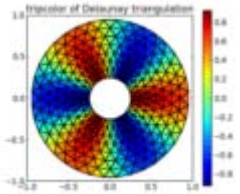


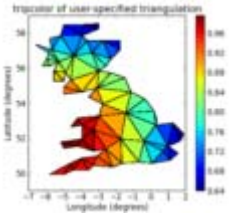


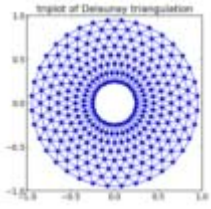

Contour plot of Delaunay triangulation

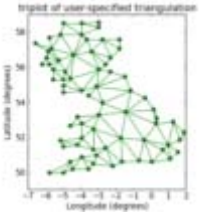

Contour plot of user-specified triangulation
Latitude (degrees)
Longitude (degrees)

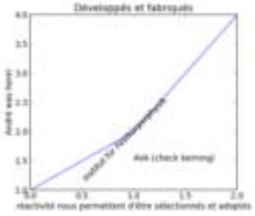

plotting and contour (200 points, 20000 grid points)
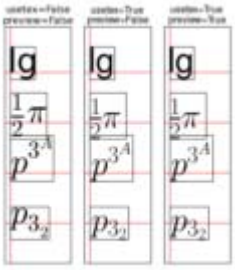

tricolor of Delaunay triangulation

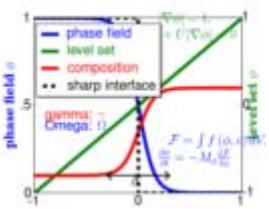

tricolor of user-specified triangulation
Latitude (degrees)
Longitude (degrees)


triplot of Delaunay triangulation

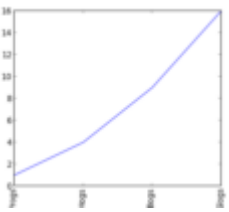

triplot of user-specified triangulation
Latitude (degrees)
Longitude (degrees)

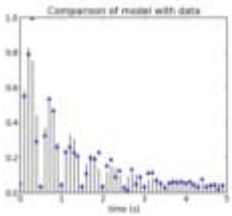

Développés et fabriqués
Axé (check kerning)
réactivité nous permettent d'être sélectionnés et adoptés

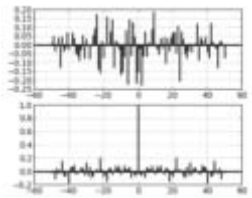

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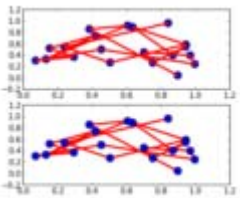

phase field
level set
composition
sharp interface
gamma:
Omega:
phase field
level set

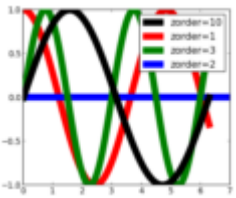

Usetex font effects

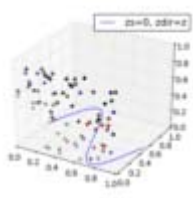


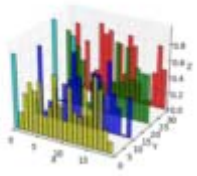
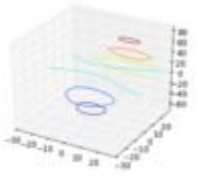
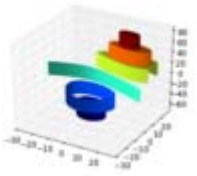
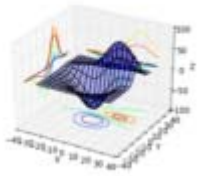
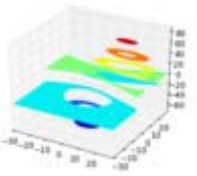
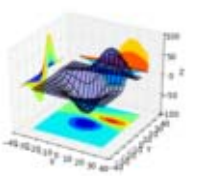
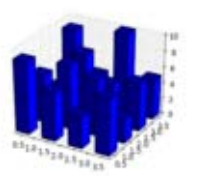
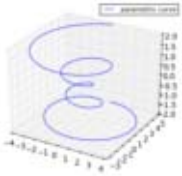
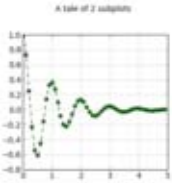
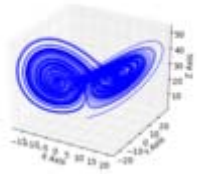
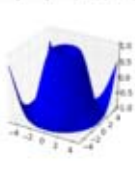
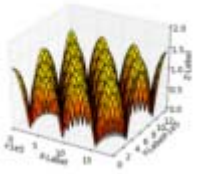
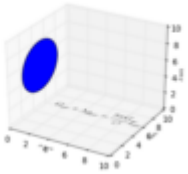
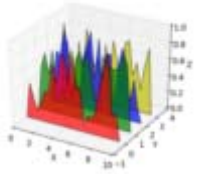
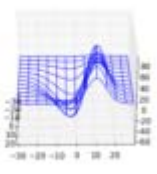
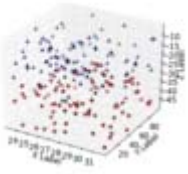
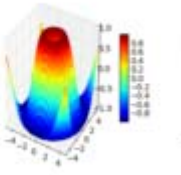
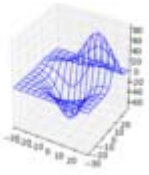
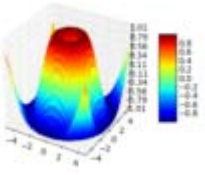
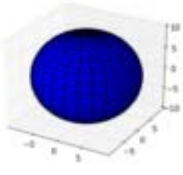
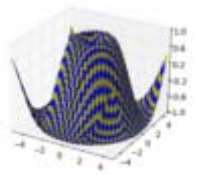
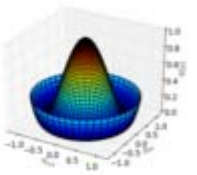
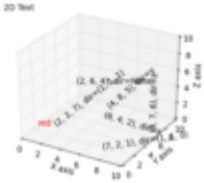
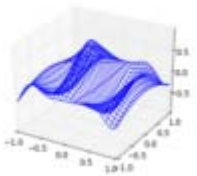

Comparison of model with data
time (s)



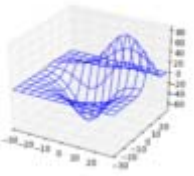


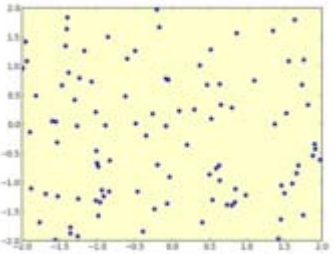

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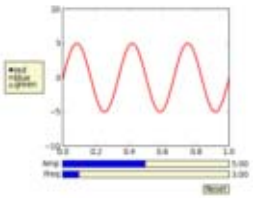

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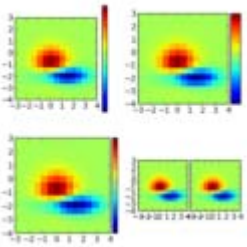









A tale of 2 subplots
















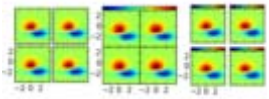
2D Text

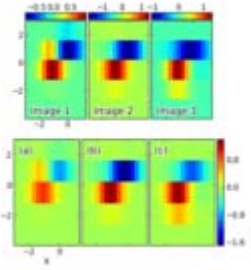


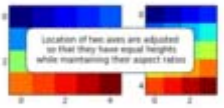


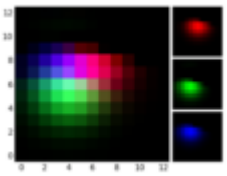


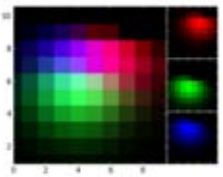


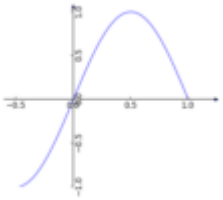


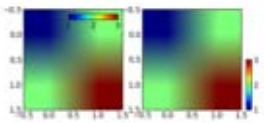

Image 1
Image 2
Image 3

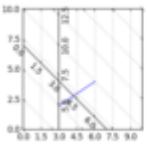

Location of two axes are adjusted
so that they have equal heights
while maintaining their aspect ratios

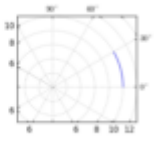


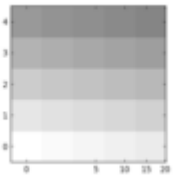


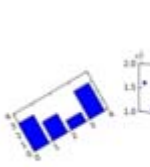


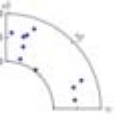




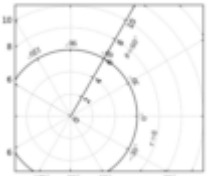


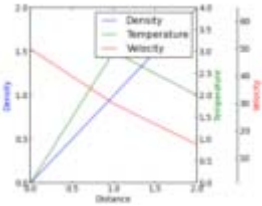


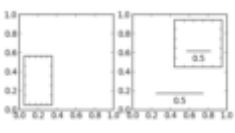


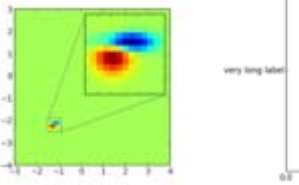


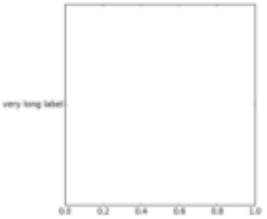


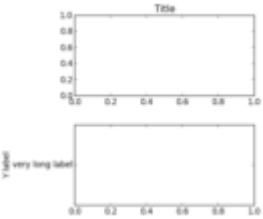


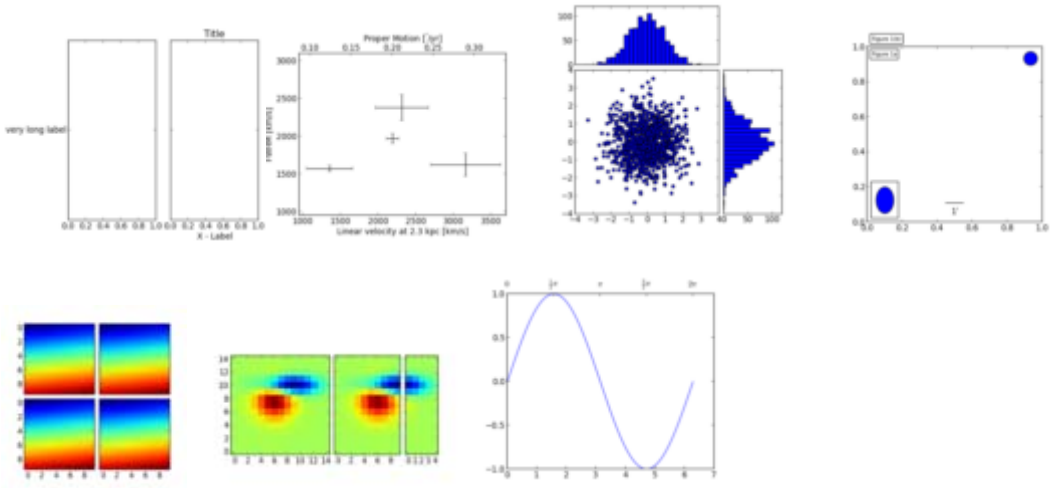

Density
Temperature
Velocity
Distance






very long label


Title
Y label
very long label


Title
very long label
X - Label
Proper Motion (''/yr)
rsense (km/s)
Linear velocity at 2.3 kpc (km/s)
Figure One
Figure Six